

LASER SPECKLE CONTRAST IMAGING PLATFORM (LSCIP)

USER MANUAL - Version 2.1

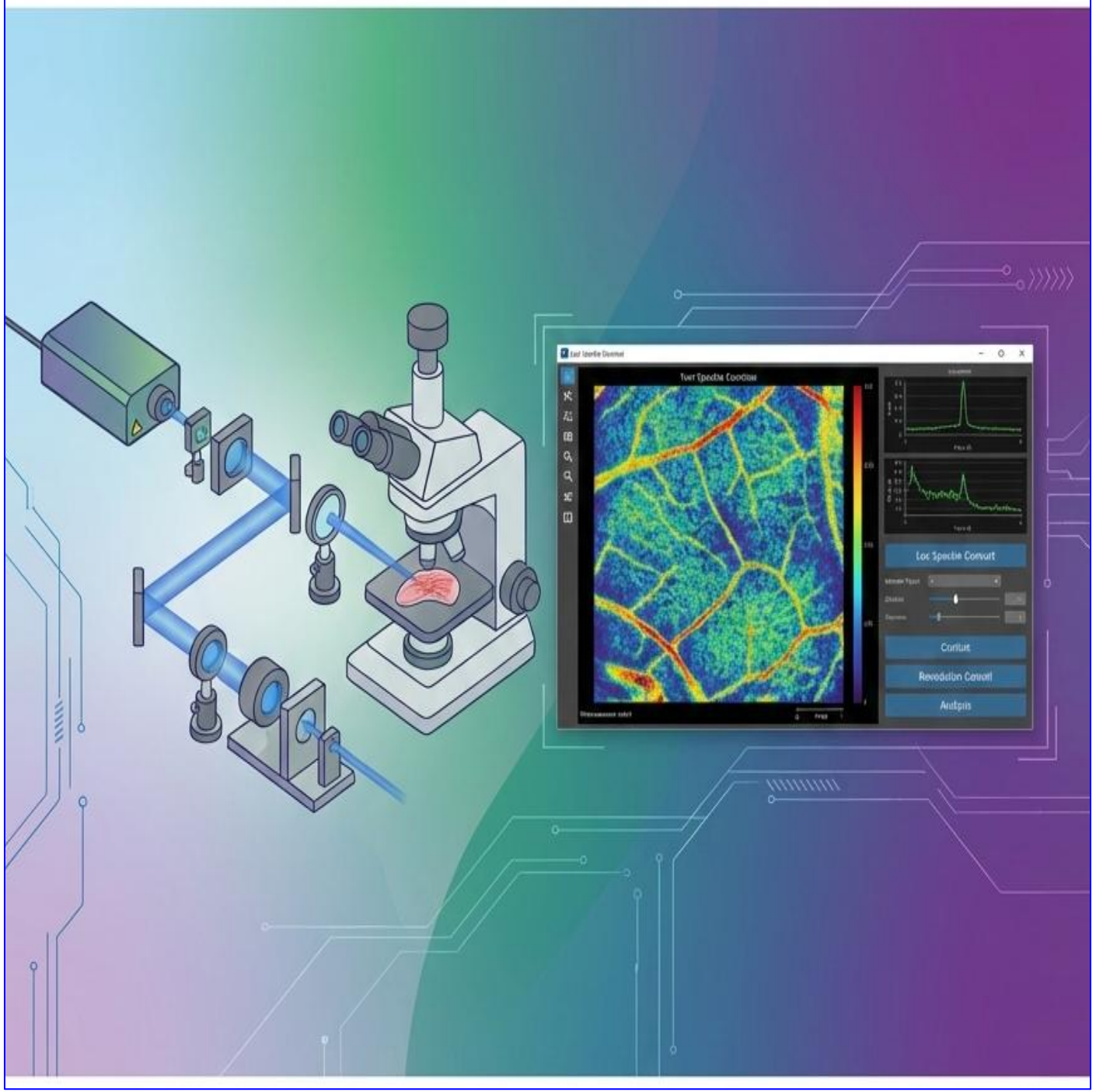


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Chapter 1 – Introduction

1.1 Purpose of the Imaging Platform

The **Laser Speckle Contrast Imaging Platform (LSCIP)** is a software-controlled experimental system designed for **high-resolution visualization and analysis of microvascular blood flow dynamics** using **Laser Speckle Contrast Imaging (LSCI)**. The platform integrates real-time camera acquisition, laser illumination control, galvo-based beam steering, synchronized stimulation triggering, and advanced image processing into a single graphical user interface (GUI).

The primary purpose of this platform is to enable **repeatable, quantitative, and spatially controlled imaging experiments**, particularly in **preclinical neuroscience, vascular physiology, and biomedical optics research**. The system allows users to:

- Acquire live and processed speckle contrast images
- Steer a laser beam across predefined spatial grids using galvo mirrors
- Perform precise pixel-to-galvo coordinate calibration
- Collect multi-exposure and multi-condition datasets automatically
- Analyze spatial and temporal intensity variations
- Synchronize imaging with external stimulation hardware

This manual provides a **complete operational reference** for users, describing not only how to operate the software, but also how the system behaves, how data are generated and stored, and how to use the platform safely and effectively.

1.2 Scientific Background: Laser Speckle Contrast Imaging

Laser Speckle Contrast Imaging is a non-contact optical technique widely used to visualize **relative blood flow and perfusion changes** in biological tissue. When coherent laser light illuminates a scattering medium such as tissue, it produces a random interference pattern known as **speckle**. The temporal fluctuations of this speckle pattern are influenced by the motion of scatterers (e.g., red blood cells).

The platform computes **speckle contrast**, typically defined as:

$$K = \frac{\sigma}{\mu}$$

where:

- σ is the standard deviation of pixel intensity
- μ is the mean pixel intensity

By calculating speckle contrast over multiple frames or spatial neighborhoods, the system provides a **relative measure of blood flow dynamics**. Lower speckle contrast generally corresponds to higher flow velocity, while higher contrast indicates slower or static flow.

This imaging platform supports:

- **Temporal speckle contrast** (multi-frame based)
- **Real-time speckle visualization**
- **Multi-exposure acquisition** for flow sensitivity analysis

The software abstracts these computations so that users can focus on **experimental design and interpretation**, rather than implementation details.

1.3 System Capabilities at a Glance

The platform combines multiple subsystems into a unified environment. Key capabilities include:

1.3.1 Imaging Capabilities

- Live grayscale camera preview
- Real-time speckle contrast video
- Adjustable exposure times (microsecond precision)
- 12-bit sensor data handling
- Region-of-Interest (ROI) based acquisition
- Pixel-to-millimeter scale calibration

1.3.2 Galvo-Based Spatial Control

- Independent X and Y galvo mirror control
- Angle-based beam steering
- Grid-based scanning patterns
- Hybrid calibration (manual + automated)
- Regression-based coordinate mapping

1.3.3 Automated Data Collection

- Multi-capture workflows
- Logarithmic exposure generation
- Dark, baseline, and stimulation datasets
- Automated directory creation and logging
- Coordinate and metadata documentation

1.3.4 Analysis and Visualization

- Intensity profile extraction
- Histogram analysis
- ROI-aware visualization
- Contrast scaling and colormap control
- Export of plots, statistics, and raw data

1.3.5 External Hardware Integration

- Laser power monitoring and control
 - Relay-based peripheral switching
 - Z-axis / stereotaxic stage positioning
 - NI-DAQ based stimulation triggering
-

1.4 Intended Users and Skill Levels

This platform is designed for **research and laboratory use**. It is not intended for clinical diagnosis or commercial deployment without appropriate validation.

Primary User Groups

1. **Graduate Students and Researchers**
 - Conduct routine imaging experiments
 - Use predefined workflows
 - Perform data analysis and export
2. **Advanced Operators / Lab Managers**
 - Perform calibration and system setup
 - Manage backups and configuration files
 - Troubleshoot hardware and software issues
3. **Developers and System Integrators**
 - Modify or extend the software
 - Integrate additional hardware modules
 - Validate system performance

Assumed Knowledge

The user manual assumes basic familiarity with:

- Optical imaging concepts
- Computer-based data acquisition
- Laboratory safety procedures
- Windows-based software operation

Advanced sections (e.g., calibration mathematics, data formats) are written to be **informative but non-mandatory** for routine operation.

1.5 Scope and Structure of This Manual

This manual is **task-oriented and system-accurate**, meaning it describes what the user sees and does, rather than how the code is implemented internally.

What This Manual Covers

- Installation and startup procedures
- GUI layout and navigation
- Imaging and speckle acquisition workflows
- Galvo calibration and scanning
- Data storage and export
- Safety guidelines and troubleshooting

What This Manual Does Not Cover

- Internal software source code
- Hardware manufacturing details
- Theoretical derivations beyond operational relevance
- Clinical interpretation of imaging results

Each chapter is structured to answer three core questions:

1. **What does this feature do?**
2. **When and why should it be used?**
3. **How does the user operate it correctly and safely?**

1.6 Versioning, Terminology, and Conventions

- The software may be referred to as **Laser Speckle Contrast Imaging Platform**, **Small Animal Cerebral Blood Flow Imaging Platform** or depending on context.
- All parameter units (μ s, mm, degrees) are explicitly stated.
- GUI button names are written **exactly as displayed** in the software.
- Warnings and safety-critical notes are clearly highlighted.

This manual corresponds to the **current stable implementation** of the GUI and should be updated whenever significant changes are made to the software or hardware configuration.

End of Chapter 1

Chapter 2 – System Overview

2.1 Overview of the Imaging System Architecture

The Speckle Contrast Imaging & Galvo-Based Imaging Platform is a **modular, software-driven experimental system** that integrates optical imaging, motion control, stimulation triggering, and data analysis into a unified framework. The system is designed around a **central host computer** running the GUI, which coordinates multiple hardware subsystems through serial, USB, and DAQ interfaces.

At a high level, the system operates as follows:

1. A high-speed camera acquires raw intensity images of the illuminated sample.
2. Coherent laser illumination produces speckle patterns sensitive to motion within the sample.
3. Galvo mirrors steer the laser beam across spatial locations.
4. The software computes speckle contrast and intensity-based metrics in real time.
5. External devices (relays, motors, DAQ stimulators) are synchronized with imaging.
6. All acquired data, metadata, and calibration results are automatically organized and stored.

This architecture allows the system to support both **interactive, real-time imaging** and **fully automated experimental workflows**.

2.2 Hardware Subsystems

The platform consists of several tightly integrated hardware components. Each subsystem is controlled independently by the software but coordinated through shared timing and configuration logic.

2.2.1 Imaging Camera Subsystem

The imaging camera is the core sensor of the system and is responsible for capturing raw intensity frames used for both live preview and speckle contrast computation.

Key characteristics:

- High-speed CMOS sensor
- 12-bit grayscale output
- Microsecond-scale exposure control
- USB-based data transfer
- External SDK-based control

Functional roles:

- Live imaging preview for alignment and focus
- Multi-frame acquisition for speckle contrast computation
- ROI-restricted acquisition for performance optimization
- Source data for intensity profiles and histograms

The camera operates in a continuous acquisition mode during live preview and switches to controlled acquisition sequences during speckle video or multi-capture workflows.

2.2.2 Laser - Illumination System

The laser subsystem provides coherent illumination necessary for speckle formation. The system assumes the laser is externally mounted and aligned with the camera field of view.

Key features:

- Continuous-wave coherent illumination
- External power supply
- Software-monitored input current
- Relay-based on/off control

Safety considerations:

- Laser current is continuously monitored and displayed in the GUI
- The software does not directly override hardware-level laser safety limits
- Operators are responsible for appropriate laser safety procedures

The laser remains active during live preview and capture sequences unless explicitly disabled by the user or via relay control.

2.2.3 Galvo Scanning System

The galvo subsystem enables **rapid, precise steering of the laser beam** across the imaging field.

Components:

- Dual-axis galvo mirrors (X and Y)
- Motor drivers controlled via serial communication
- Angle-based positioning interface

Capabilities:

- Continuous angular positioning ($\pm$ range defined by hardware)
- Grid-based scanning patterns

- Manual and automated movement
- Integration with calibration mapping

The galvo system is central to spatially resolved measurements, enabling multi-point sampling without physically moving the sample or camera.

2.2.4 Relay and Peripheral Control System

A relay module provides software-controlled switching of auxiliary devices.

Typical uses include:

- Laser enable/disable
- External illumination control
- Peripheral device switching

Characteristics:

- Multi-channel relay module
- Shared serial communication interface
- Visual ON/OFF status feedback in GUI

Relay states are logged implicitly as part of the experimental context and should be managed carefully during automated acquisition sequences.

2.2.5 Z-Axis / Stereotaxic Stage System

The platform supports a motorized Z-axis or stereotaxic stage for fine positional adjustment.

Functions:

- Focus optimization
- Sample positioning
- Depth-dependent imaging

Control method:

- Serial or Modbus-based communication
- Incremental movement commands
- Integrated into the camera control window

This subsystem allows precise adjustment without disturbing lateral alignment or galvo calibration.

2.2.6 NI-DAQ Stimulation Subsystem

An optional NI-DAQ device enables synchronized stimulation during imaging experiments.

Capabilities:

- Digital pulse generation
- User-configurable pulse timing
- Integration with multi-capture workflows

Typical applications:

- Forepaw stimulation
- Electrical or optical triggering
- Time-locked experimental paradigms

The stimulation subsystem is designed to operate in coordination with image acquisition, ensuring reproducible timing across trials.

2.3 Software Architecture Overview

The software is implemented as a **Python-based GUI application** running on Windows. It uses a combination of:

- GUI frameworks for user interaction
- Hardware SDKs for device control
- Numerical libraries for image processing
- File management utilities for structured data storage

Key architectural principles:

- Separation of hardware control and user interaction
- Modular tool windows for specialized tasks
- Background processing for continuous acquisition
- Automatic recovery and status reporting where possible

The software maintains internal state awareness, including camera status, calibration validity, ROI configuration, and galvo enable state.

2.4 Data Flow and Processing Pipeline

The system processes data through a structured pipeline:

Acquisition → **Processing**(Pre-processing, Computation, Visualization) → **Storage**

1. **Acquisition**
Raw frames are acquired from the camera at the configured exposure and frame rate.
2. **Preprocessing**
Frames are optionally cropped to the ROI and scaled according to bit depth.
3. **Computation**
 - Speckle contrast is computed using multi-frame statistics
 - Intensity profiles and histograms are derived as needed
4. **Visualization**
Results are displayed in real time with adjustable scaling and colormaps.
5. **Storage**
Data are saved in structured directories with associated metadata, logs, and previews.

This pipeline supports both **real-time visualization** and **offline quantitative analysis**.

2.5 Operating Modes

The platform supports multiple operating modes, each optimized for a specific task.

2.5.1 Live Imaging Mode

- Continuous camera preview
- Manual adjustment of exposure and focus
- Alignment and ROI selection

2.5.2 Speckle Contrast Video Mode

- Real-time speckle computation
- Adjustable frame aggregation
- Visual flow contrast assessment

2.5.3 Galvo Scanning Mode

- Sequential beam positioning
- Grid-based sampling
- Position logging and repeatability

2.5.4 Multi-Capture Experimental Mode

- Automated exposure sweeps
- Multiple data types (dark, baseline, stimulation)
- Fully unattended acquisition sequences

Each mode shares the same underlying hardware but applies different control logic and data handling rules.

2.6 Calibration Dependencies and System State

Accurate system operation depends on several calibration layers:

- Pixel-to-millimeter scale calibration
- Galvo angle-to-pixel mapping
- ROI-dependent coordinate transformation

The software continuously tracks whether valid calibration data are available and uses this information to enable or restrict certain actions.

Changes to ROI, camera resolution, or optical alignment may invalidate existing calibration and require recalibration.

2.7 Design Philosophy and Extensibility

The system is designed with extensibility in mind:

- New hardware modules can be integrated via additional control windows
- Analysis tools can be expanded without disrupting acquisition logic
- Calibration formats support both human-readable and binary storage

This design allows the platform to evolve alongside experimental requirements while preserving backward compatibility with existing datasets.

2.8 Summary

This chapter has described the **complete hardware–software ecosystem** of the Speckle Contrast Imaging Platform. Understanding this system-level structure is essential before performing installation, calibration, or complex experiments.

Subsequent chapters will build on this foundation, moving from conceptual understanding to **hands-on operation and experimental workflows**.

End of Chapter 2

Chapter 3 – System Requirements and Installation

3.1 Introduction

This chapter describes the **hardware, software, and environmental requirements** necessary for correct operation of the Speckle Contrast Imaging & Galvo-Based Imaging Platform, followed by **step-by-step installation and initial setup procedures**.

Correct installation is critical. Many system functions—including camera acquisition, galvo calibration, speckle computation, and stimulation synchronization—depend on precise driver versions, communication ports, and startup order. Improper installation may result in partial system operation that is difficult to diagnose.

This chapter should be completed **before attempting to launch the GUI for the first time**.

3.2 Hardware Requirements

3.2.1 Host Computer Requirements

The system is designed to run on a dedicated Windows-based workstation.

Minimum recommended specifications:

- Operating System: Windows 10 or Windows 11 (64-bit)
- Processor: Quad-core CPU (Intel i5 / AMD Ryzen 5 or higher)
- RAM: 16 GB (32 GB recommended for high-frame-rate operation)
- Storage:
 - SSD required for system drive
 - ≥ 500 GB free space recommended for experimental data
- USB Ports:
 - At least one high-speed USB port for camera
 - Additional USB ports for serial devices
- Display:
 - Minimum resolution: 1920×1080
 - Dual-monitor setup recommended for analysis workflows

The system should not be shared with unrelated high-load software during experiments.

3.2.2 Imaging Camera

The imaging camera is a **high-speed grayscale sensor** compatible with the scTDC camera SDK.(ReconFlex 1920 Camera has been used here)

Key requirements:

- 12-bit output support
- External SDK and driver installation
- Stable USB data connection
- Dedicated power supply (if applicable)

The camera must be connected **only after the appropriate drivers and SDK have been installed.**

3.2.3 Laser Illumination Hardware

The laser subsystem is externally powered and aligned.

Requirements:

- Coherent laser source compatible with speckle imaging
- Stable optical mount
- External power/current control
- Laser safety interlock (hardware-level)

The software monitors laser input current but does **not replace physical laser safety systems.**

3.2.4 Galvo Mirror and Driver System

The galvo subsystem enables beam steering and spatial scanning.

Components:

- Dual-axis galvo mirrors (X and Y)
- Motor driver/controller
- Serial or USB communication interface

Requirements:

- Stable power supply
- Known angular range limits
- Shared communication port with relay or motor controller (if applicable)

3.2.5 Relay Module

A multi-channel relay board provides software-controlled switching.

Typical uses:

- Laser enable/disable
- Peripheral device switching

The relay module communicates via serial interface and is automatically detected at startup.

3.2.6 Z-Axis / Stereotaxic Stage

If installed, the Z-axis or stereotaxic stage allows vertical positioning.

Requirements:

- Motorized stage with serial or Modbus control
 - Known movement increments
 - Stable mechanical mounting
-

3.2.7 NI-DAQ Device (Optional)

For stimulation experiments, an NI-DAQ device is required.

Requirements:

- National Instruments DAQ hardware
- Digital output capability
- Compatible NI-DAQmx drivers

This subsystem is optional but fully integrated into the software when present.

3.3 Software Requirements

3.3.1 Operating System

- Windows 10 or Windows 11 (64-bit)
 - Administrator access required for driver installation
 - Power management disabled for USB devices
-

3.3.2 Python Environment

The GUI is written in Python and requires a compatible Python installation.

Recommended configuration:

- Python version: 3.8 or later
 - 64-bit Python distribution
 - Virtual environment recommended but not mandatory
-

3.3.3 Required Python Libraries

The following categories of libraries are required:

Core libraries

- numpy
- scipy
- matplotlib
- pillow (PIL)

GUI libraries

- tkinter
- customtkinter (if used)

Image processing

- opencv-python (cv2)

Hardware communication

- pyserial
- pymodbus
- PyDAQmx (for NI-DAQ systems)

Camera SDK

- scTDC Python bindings

All libraries must be installed **before launching the GUI**.

3.3.4 Device Drivers and SDKs

The following drivers must be installed **in the correct order**:

1. Camera driver and SDK (from manufacturer)
2. NI-DAQmx drivers (if NI-DAQ is used)

3. USB-to-serial drivers (if required by relay/galvo controllers)

After installation, the system should be rebooted.

3.4 Physical Setup and Cabling

3.4.1 Recommended Power-On Order

1. Ensure all devices are powered OFF
2. Power on relay, galvo, and motor controllers
3. Power on laser (standby mode)
4. Connect camera via USB
5. Power on host computer

This sequence reduces the likelihood of communication failures.

3.4.2 USB and Serial Connections

- Use short, shielded USB cables
- Avoid USB hubs for the camera
- Assign consistent COM ports when possible
- Label cables and ports physically

3.4.3 Optical Alignment Considerations

- Camera field of view must overlap laser illumination
- Galvo zero-angle position should align near image center
- Focus should be approximately set before calibration

3.5 Directory Structure and File Organization

The software expects a consistent directory layout.

Typical structure:

```
GUI_Project/  
├── LSCIP_GUI.py  
├── calibration/  
├── backups/  
├── logs/  
├── data/  
└── config/
```

└─ drivers/

Calibration and backup files are automatically generated during operation.

3.9 Safety and Compliance Notes

- Always verify laser safety compliance before operation
 - Do not bypass hardware interlocks
 - Ensure stimulation parameters comply with institutional protocols
 - Maintain physical emergency stop access
-

3.10 Summary

This chapter has detailed all **hardware, software, and installation requirements** necessary to bring the system to a functional state. Completing these steps ensures that subsequent chapters—covering GUI operation, imaging, calibration, and experiments—can be followed reliably and reproducibly.

End of Chapter 3

Chapter 4 – Starting the Application

4.1 Purpose of This Chapter

This chapter describes **how the software application is started**, what occurs internally during startup, how to interpret startup messages, and how to confirm that the system has entered a **safe and ready operational state**.

Unlike later chapters that focus on imaging and experimentation, this chapter emphasizes:

- Correct startup order
- Automatic detection and initialization logic
- Early identification of hardware or configuration problems
- User responsibilities during the startup phase

A disciplined startup procedure significantly reduces downstream errors related to calibration, communication timeouts, and incomplete data acquisition.

4.2 Pre-Launch Checklist

Before starting the application, confirm the following conditions:

4.2.1 Hardware Power State

- Camera is powered and connected via USB
- Galvo controller and relay modules are powered
- Z-axis or stereotaxic stage (if used) is powered
- NI-DAQ device (if used) is connected and powered
- Laser is powered but set to **safe standby output**

4.2.2 Physical and Environmental Checks

- Optical components are securely mounted
- No loose cables are present
- Emergency stop access is unobstructed
- Laser safety eyewear is available and worn if required

4.2.3 Software Readiness

- Correct Python environment is available
- Required drivers and SDKs are installed
- No other software is currently accessing the camera or DAQ device

Only after these checks are satisfied should the application be launched.

4.3 Launching the GUI

4.3.1 Standard Launch Method

The application is launched by running the main GUI script:

```
new_GUI_sc_55.py
```

This may be done via:

- Command Prompt
- PowerShell
- Python IDE (e.g., VS Code, PyCharm)
- Double-click execution (if file associations are configured)

It is recommended to launch from a terminal so that **console messages remain visible**, especially during troubleshooting.

4.3.2 What Happens Immediately After Launch

Once the script is executed, the following sequence occurs automatically:

1. **GUI Framework Initialization**
 - The main application window is created
 - Visual layout elements are instantiated
 - Default styles and fonts are applied
2. **Internal State Initialization**
 - Software variables are reset to default values
 - Flags indicating camera, galvo, and calibration status are cleared
 - Default configuration values are loaded
3. **Background Threads Prepared**
 - Acquisition and processing threads are prepared but not started
 - Event handlers are registered

At this stage, the GUI window becomes visible to the user.

4.4 Automatic Hardware Detection

After the GUI becomes visible, the application begins **automatic hardware discovery**. This process is essential for seamless operation and minimizes the need for manual configuration.

4.4.1 Serial and COM Port Detection

The application scans available serial ports to identify connected devices such as:

- Galvo controller
- Relay module
- Z-axis motor controller
- Laser current monitor

Detection behavior:

- Known device signatures are matched
- Ports are assigned internally
- Non-critical devices may be skipped if absent

If a device is not detected, the application:

- Logs the failure internally
- Displays a warning (if applicable)
- Continues launching unless the device is mandatory

4.4.2 Camera Discovery

The camera is detected using the manufacturer-provided SDK.

During this step:

- The SDK is queried for available devices
- Camera capabilities are read
- Resolution and bit-depth settings are validated

The camera is **not yet acquiring images** at this stage. Acquisition begins only after explicit user action.

4.4.3 NI-DAQ Device Detection (Optional)

If NI-DAQ functionality is enabled:

- The application queries installed NI-DAQ devices
- Device names and channels are verified
- Stimulation parameters are initialized

Failure to detect an NI-DAQ device does not prevent imaging but disables stimulation features.

4.5 Calibration Auto-Loading

Once hardware detection is complete, the software attempts to load previously saved calibration data.

4.5.1 Galvo Calibration Files

The application searches predefined locations for:

- Active galvo calibration files
- Backup calibration files

If a valid calibration is found:

- The calibration matrix is loaded into memory
- Calibration status is set to “valid”
- Galvo-based features become available

If no calibration is found:

- The system remains operational
 - Galvo scanning and coordinate mapping are disabled
 - The user is expected to perform calibration manually
-

4.5.2 Pixel-to-Millimeter Scale Calibration

If a pixel-to-millimeter calibration factor exists:

- The factor is loaded automatically
- On-screen rulers and measurements are updated

If absent:

- Measurements default to pixel units
 - Scale calibration tools remain accessible
-

4.6 Initial GUI State After Startup

When startup is complete, the GUI enters a **ready but idle state**.

4.6.1 Expected Visual Indicators

- Main window fully rendered
- Live image panel empty or black
- Speckle panel inactive
- Status indicators show “Camera not initialized” or equivalent
- Laser current display shows live readings (if connected)

4.6.2 Disabled vs Enabled Controls

- Camera initialization button is enabled
- Preview and capture buttons are disabled
- Galvo enable toggle may be disabled until camera is initialized
- Analysis tools are unavailable until imaging begins

This state is intentional and prevents unsafe or invalid operations.

4.7 User Actions Required After Startup

After the application has launched successfully, the user must perform the following actions in order:

1. **Initialize the Camera**
2. **Start Live Preview**
3. **Adjust Exposure and Focus**
4. **Verify Laser Illumination**
5. **Confirm Galvo Response (if used)**

These steps are covered in detail in later chapters but are listed here to clarify that **startup alone does not begin imaging**.

4.8 Startup Messages and Status Feedback

Throughout the startup process, the application provides feedback via:

- Status bar messages
- Console output
- Pop-up warnings (when critical)

Users should become familiar with common messages such as:

- “Camera initialized successfully”
- “Calibration loaded”

- “Device not detected – feature disabled”

These messages provide early insight into system readiness.

4.9 Common Startup Issues and Immediate Remedies

4.9.1 GUI Opens but Camera Cannot Be Initialized

- Camera SDK not installed correctly
- Camera in use by another application
- USB bandwidth issue

4.9.2 Serial Devices Not Detected

- Device powered off
- Incorrect COM port assignment
- USB-to-serial driver missing

4.9.3 Calibration Not Loaded

- Calibration files missing or corrupted
- ROI mismatch with previous calibration
- First-time system use

These issues should be resolved **before proceeding to imaging or calibration**.

4.10 Safe Shutdown and Restart

If a critical issue is detected during startup:

1. Close the GUI using the Exit button
2. Power down external hardware if necessary
3. Resolve the issue
4. Restart the application following the same procedure

Abrupt termination during startup should be avoided, especially if hardware is actively communicating.

4.11 Summary

This chapter has described the **complete startup lifecycle** of the application, from pre-launch checks to a verified ready state. Understanding this process ensures that all subsequent imaging, calibration, and experimental steps are performed on a stable and predictable foundation.

End of Chapter 4

Chapter 5 – Graphical User Interface (GUI) Layout and Navigation

5.1 Purpose of the GUI

The Graphical User Interface (GUI) is the **primary point of interaction** between the user and the imaging system. It is designed to coordinate multiple hardware subsystems, real-time image acquisition, processing pipelines, and data management workflows through a **single, unified control surface**.

The GUI follows three core design principles:

1. **State Awareness**
Controls are enabled or disabled based on the current system state (e.g., camera initialized, calibration loaded).
2. **Task-Oriented Grouping**
Controls are arranged by workflow: imaging, calibration, analysis, and automation.
3. **Safety and Error Prevention**
Unsafe or invalid actions are restricted until prerequisites are satisfied.

This chapter explains the GUI layout **region by region**, describing not only what each control does, but also **when it becomes active, what dependencies it has, and how it interacts with other parts of the system**.

5.2 Overall Window Layout

The GUI window is divided into five major regions:

1. **Header Bar (Top)**
2. **Quick Actions Panel (Left)**
3. **Live Image Display Panel (Center-Left)**
4. **Speckle Contrast Display Panel (Center-Right)**
5. **Status and Feedback Region (Bottom / Embedded)**

Each region serves a distinct operational purpose, and together they represent the complete experimental control interface.

5.3 Header Bar

The header bar spans the top of the GUI and provides **global system controls and indicators** that remain visible at all times.

5.3.1 System Identification and Status

The header typically displays:

- System name and version
- Active session context
- High-level status indicators

This allows the user to immediately confirm that the correct software instance is running.

5.3.2 Laser Current Monitor

A prominent element of the header bar is the **real-time laser input current display**.

Purpose:

- Provides continuous feedback on laser operating conditions
- Acts as a safety awareness indicator

Behavior:

- Updates automatically when the laser monitor is connected
- Displays numeric current values (e.g., in mA)
- Remains visible regardless of imaging state

The software **does not override hardware limits**, but this display helps users detect abnormal behavior early.

5.3.3 Global Action Buttons

The header includes buttons that open **system-wide tools and dialogs**, such as:

- Scale calibration
- Relay control
- NI-DAQ stimulation control
- Help, specifications, and about dialogs
- Application exit

These controls are intentionally separated from imaging actions to reduce accidental activation during experiments.

5.4 Quick Actions Panel (Left Panel)

The Quick Actions Panel is the **primary operational control area**. It contains vertically arranged buttons grouped by workflow stage.

This panel guides the user through a **top-to-bottom operational sequence**, from hardware initialization to advanced analysis.

5.4.1 Camera Control Actions

These buttons manage camera state and access camera-related settings.

Initialize Camera

- Establishes communication with the camera
- Allocates acquisition buffers
- Must be pressed before any imaging is possible

This button is enabled immediately after startup and disabled once the camera is successfully initialized.

Open Camera Controls

- Opens a dedicated control window
- Provides access to:
 - Exposure time
 - ROI settings
 - Z-axis control
 - Scale calibration factor

This window acts as the **low-level camera configuration interface**.

Start Preview / Stop Preview

- Starts or stops continuous live acquisition
- Updates the Live Image Display panel
- Enables downstream tools once active

Preview must be stopped before certain operations (e.g., shutdown or some calibration steps).

5.4.2 Motion and Platform Controls

Stereotaxic Platform / Motor Control

- Opens the motor control window
- Allows fine Z-axis or platform adjustment
- Typically used during focusing and alignment

Movement commands are incremental to reduce mechanical risk.

5.4.3 Multi-Capture and Galvo Control

Multi-Capture & Galvo

- Opens a multi-tab control window
- Central hub for:
 - Galvo angle control
 - Galvo calibration
 - Grid scanning
 - Automated multi-capture experiments

This is one of the most complex GUI components and is intentionally isolated to avoid accidental use.

5.4.4 Advanced Imaging and Analysis Launchers

Spatial-Temporal Contrast

- Opens tools for advanced contrast computation
- Used for extended or experimental analysis modes

Launch Speckle Tracker

- Launches an external or auxiliary speckle analysis tool
 - Intended for post-processing or tracking applications
-

5.4.5 Speckle Contrast Video Controls

A dedicated subsection within the Quick Actions panel controls real-time speckle video.

Controls include:

- Number of frames per speckle computation

- Start speckle video
- Stop speckle video

These controls remain disabled until live preview is active.

5.4.6 Intensity Profile and Image Saving Tools

This subsection provides quick access to analysis and export tools:

- Live intensity profile
- Full live image histogram
- Speckle intensity profile
- Full speckle histogram
- Save live image
- Save speckle image

These tools depend on active image data and are dynamically enabled or disabled.

5.4.7 Feed Visibility Toggles

Checkboxes allow the user to:

- Show or hide live feed
- Show or hide speckle feed

This is useful when focusing on a single visualization mode or optimizing performance.

5.5 Live Image Display Panel

The Live Image Display Panel presents the **raw grayscale camera feed**.

5.5.1 Display Characteristics

- Displays 12-bit image data scaled for visualization
- Supports ROI-based cropping
- Updates continuously during preview

This panel is primarily used for:

- Alignment
- Focusing

- ROI definition
 - Calibration interaction
-

5.5.2 Cursor and Coordinate Feedback

When the mouse moves over the live image:

- Pixel coordinates are displayed
 - ROI-relative and absolute coordinates may be shown
 - Real-time feedback aids calibration and profile selection
-

5.5.3 Scale and Rulers

If pixel-to-millimeter calibration is available:

- On-screen rulers are displayed
- Measurements reflect real-world dimensions

If calibration is absent, measurements default to pixels.

5.6 Speckle Contrast Display Panel

The Speckle Contrast Display Panel shows **processed speckle contrast images**.

5.6.1 Visualization Characteristics

- Displays computed speckle contrast maps
 - Uses false-color colormaps
 - Includes a colorbar indicating contrast scale
-

5.6.2 Contrast Scaling (C-Axis Control)

A vertical slider allows adjustment of the contrast display range.

Important characteristics:

- Does not recompute speckle data
- Only affects visualization scaling
- Allows enhancement of subtle flow differences

5.6.3 Interaction with Analysis Tools

The speckle display is interactive:

- Supports intensity profile drawing
- Supports histogram analysis
- Reflects ROI settings

5.7 Status Indicators and Feedback

Throughout the GUI, the user receives continuous feedback via:

- Status messages
- Progress bars during capture
- Enable/disable states of controls
- Warning dialogs

These indicators provide essential context for understanding what the system is currently doing and what actions are allowed.

5.8 GUI State Dependencies

The GUI enforces a strict dependency hierarchy:

- Camera must be initialized before preview
- Preview must be active before speckle video
- Calibration must be valid before galvo scanning
- ROI changes may invalidate calibration
- Data capture requires valid save directory

Understanding these dependencies prevents misuse and data corruption.

5.9 Navigation Best Practices

- Work **top to bottom** in the Quick Actions panel
 - Confirm status messages before proceeding
 - Avoid opening multiple control windows unnecessarily
 - Stop preview before exiting or recalibrating hardware
-

5.10 Summary

This chapter has described the **full layout, logic, and navigation principles of the GUI**. Mastery of the interface is essential for safe, efficient, and reproducible operation of the imaging system.

The next chapters transition from interface familiarity to **hands-on operation**, beginning with live imaging and preview workflows.

End of Chapter 5

Chapter 6 – Getting Started: Basic Imaging Workflow

6.1 Power-On and Connection Checklist

Before initiating any imaging operation, it is essential to ensure that **all hardware components are correctly powered, connected, and in a safe state**. This checklist should be followed every time the system is used, particularly after shutdown, relocation, or maintenance.

6.1.1 Hardware Power Checklist

Verify the following devices are powered ON:

- Host computer (Windows system)
- Imaging camera (if externally powered)
- Galvo controller and drivers
- Relay module
- Z-axis or stereotaxic stage (if installed)
- NI-DAQ device (if used)
- Laser power supply (set to safe standby level)

The laser should be **enabled electrically only if required**, and optical output should be verified to be within safe operating limits.

6.1.2 Physical Connections

Confirm all physical connections:

- Camera securely connected via USB (preferably direct, no hub)
- Serial/USB connections to:
 - Galvo controller
 - Relay module
 - Z-axis controller
- NI-DAQ connected via USB or PCI interface
- No loose or strained cables

Connections should remain unchanged during operation to avoid communication loss.

6.1.3 Optical and Mechanical Readiness

- Camera lens or objective is securely mounted
 - Laser beam path is unobstructed
 - Sample is positioned securely
 - Z-axis has sufficient travel range in both directions
 - No mechanical interference with galvo mirrors or stage
-

6.1.4 Software Readiness

- Application launched successfully
- No startup error dialogs present
- Console output indicates successful hardware detection
- GUI responsive and idle (camera not yet initialized)

Only proceed once all checklist items are satisfied.

6.2 Camera Initialization

Camera initialization establishes communication between the software and the imaging sensor and allocates internal buffers required for image acquisition.

6.2.1 Purpose of Camera Initialization

Camera initialization performs the following actions:

- Detects the connected camera via SDK
- Queries sensor capabilities (resolution, bit depth)
- Allocates frame buffers
- Sets default acquisition parameters
- Locks the camera for exclusive access

No imaging operations are possible until this step is completed.

6.2.2 Initializing the Camera

To initialize the camera:

1. Locate the **“Initialize Camera”** button in the Quick Actions panel
2. Click the button once
3. Observe the status message indicating initialization progress

Successful initialization is typically indicated by:

- A confirmation message in the GUI or console
 - Activation of preview-related controls
 - Change in button states (initialize disabled)
-

6.2.3 Common Initialization Issues

- **Camera not detected:** Check USB connection and driver installation
- **Camera already in use:** Ensure no other application is accessing the camera
- **SDK error:** Restart application and power-cycle camera

Initialization should not be repeated unless the camera is explicitly released.

6.3 Starting and Stopping Live Preview

Live preview provides continuous real-time imaging and is the foundation for alignment, focusing, ROI selection, and exposure adjustment.

6.3.1 Starting Live Preview

To start live preview:

1. Ensure the camera is initialized
2. Click the **“Start Preview”** button
3. Observe the Live Image Display panel

Once preview begins:

- The camera enters continuous acquisition mode
 - The live image panel updates in real time
 - Downstream tools become enabled
-

6.3.2 Understanding Live Preview Behavior

During preview:

- Images are displayed in grayscale
- Brightness depends on exposure and illumination
- Motion appears smoothly if frame rate is sufficient

Live preview **does not automatically save data**.

6.3.3 Stopping Live Preview

To stop preview:

- Click the “**Stop Preview**” button

Stopping preview:

- Halts camera acquisition
- Releases processing resources
- Is required before some calibration or shutdown actions

Preview should always be stopped before exiting the application.

6.4 Exposure Control and Optimization

Exposure time directly affects image brightness, contrast, and speckle quality. Proper exposure is essential for meaningful speckle contrast computation.

6.4.1 Accessing Exposure Controls

Exposure controls are located in the **Camera Control Window**, accessible via the GUI.

Typical exposure units:

- Microseconds (μs) or milliseconds (ms)
-

6.4.2 Principles of Exposure Optimization

Proper exposure should satisfy the following:

- Avoid pixel saturation (clipping)
 - Maintain sufficient signal-to-noise ratio
 - Ensure stable speckle patterns
 - Use consistent exposure across experiments
-

6.4.3 Step-by-Step Exposure Adjustment

1. Start live preview
2. Set an initial moderate exposure
3. Observe image brightness
4. Increase exposure gradually until:
 - Speckle pattern is clearly visible
 - No large saturated regions appear
5. Fine-tune exposure based on histogram feedback

Histogram tools are recommended to confirm proper dynamic range usage.

6.4.4 Exposure and Speckle Imaging

- Too short exposure → noisy speckle contrast
 - Too long exposure → washed-out or saturated image
 - Exposure should remain fixed during comparative experiments
-

6.5 Focus and Z-Axis Adjustment

Accurate focus is critical for both raw imaging and speckle contrast quality.

6.5.1 Manual Focus Adjustment

If using a manual lens or objective:

- Adjust focus ring slowly during live preview
 - Look for maximum speckle sharpness
 - Avoid excessive motion during adjustment
-

6.5.2 Z-Axis Motorized Adjustment

If a Z-axis stage is available:

1. Open the Z-axis or platform control window
2. Use small incremental movements
3. Observe live image changes after each step

Focus should be optimized for:

- Maximum speckle visibility
- Sharp sample features
- Stable image intensity

6.5.3 Best Practices for Focus

- Always focus under final exposure conditions
 - Refocus if ROI or sample position changes
 - Avoid refocusing after calibration without revalidation
-

6.6 Typical First-Time Imaging Session

This section describes a **complete, end-to-end first imaging session**, combining all previous steps into a practical workflow.

6.6.1 Step-by-Step Example Workflow

1. Power on all hardware components
 2. Launch the GUI application
 3. Confirm no startup errors
 4. Initialize the camera
 5. Start live preview
 6. Verify laser illumination
 7. Adjust exposure for optimal brightness
 8. Focus using manual or Z-axis control
 9. Confirm image stability and clarity
 10. Stop preview (if proceeding to calibration or capture)
-

6.6.2 Expected Results

At the end of a successful first session:

- Live images appear stable and well-exposed
 - Speckle patterns are clearly visible
 - GUI controls respond correctly
 - No error or warning dialogs remain active
-

6.6.3 Common Beginner Mistakes

- Forgetting to initialize the camera
- Adjusting exposure with preview stopped
- Overexposing the image
- Attempting speckle analysis before preview

- Skipping focus optimization

6.6.4 Transition to Advanced Workflows

Once basic imaging is verified, users can proceed to:

- Speckle contrast video
- ROI selection
- Calibration procedures
- Galvo scanning
- Multi-capture experiments

These topics are covered in subsequent chapters.

6.7 Summary

This chapter has provided a **complete, practical guide to basic imaging operation**, from powering on the system to completing a first imaging session. Mastery of these steps ensures reliable performance, high-quality data, and safe system operation.

End of Chapter 6

Chapter 7 – Camera Control and Image Acquisition

7.1 Camera Initialization and Operational States

The camera subsystem operates as a **state-driven device**, meaning that only certain actions are permitted depending on the current camera state. Understanding these states is essential for correct operation and troubleshooting.

7.1.1 Camera Initialization Process

Camera initialization is a one-time process per application session and involves the following steps:

1. Discovery of connected camera hardware via the SDK
2. Verification of camera firmware and compatibility
3. Allocation of internal memory buffers for image frames
4. Configuration of default acquisition parameters
5. Establishment of exclusive access to the device

Once initialized, the camera remains reserved for the application until it is explicitly released or the application exits.

7.1.2 Camera States Overview

The camera transitions through several distinct operational states:

<i>Camera State</i>	<i>Description</i>	<i>Allowed Actions</i>
Uninitialized	Camera not yet connected or detected	Initialization only
Initialized (Idle)	Camera ready but not acquiring	Parameter changes
Preview (Continuous Acquisition)	Live imaging active	Exposure adjustment, focus
Capture (Controlled Acquisition)	Data capture in progress	Monitoring only
Error State	Communication or hardware fault	Reset or restart

The GUI dynamically enables or disables controls based on the current state.

7.1.3 State Transitions and Safety

State transitions are controlled internally to prevent invalid operations such as:

- Changing exposure during frame capture
- Starting multiple acquisition threads
- Reinitializing the camera during acquisition

Attempted invalid transitions are blocked and may generate warnings.

7.2 Exposure Time Control

Exposure time determines how long the camera sensor integrates light for each frame and is one of the most critical acquisition parameters.

7.2.1 Exposure Time Definition

Exposure time is defined as the duration during which each pixel accumulates charge in response to incident photons.

Units:

- Microseconds (μs)
 - Milliseconds (ms)
-

7.2.2 Exposure Control Interface

Exposure is controlled through the **Camera Control Window**, where the user can:

- Enter numeric exposure values
- Increment or decrement exposure in fine steps
- Apply changes in real time during preview

Exposure changes take effect immediately in live preview mode.

7.2.3 Exposure Constraints and Limits

Exposure values are constrained by:

- Camera hardware limits
- Selected resolution and frame rate

- USB bandwidth limitations

The software automatically enforces valid ranges to prevent SDK errors.

7.2.4 Exposure and Image Quality Trade-Offs

- **Short exposure:**
 - Reduced motion blur
 - Increased noise
 - Lower signal-to-noise ratio
- **Long exposure:**
 - Higher brightness
 - Risk of saturation
 - Reduced speckle contrast accuracy

Exposure should be chosen to avoid both underexposure and saturation.

7.2.5 Exposure Consistency Across Experiments

For reproducibility:

- Use identical exposure settings across comparative datasets
 - Record exposure values in experimental logs
 - Avoid dynamic exposure changes during capture sequences
-

7.3 Frame Acquisition Modes

The system supports multiple acquisition modes, each optimized for a different use case.

7.3.1 Continuous (Preview) Acquisition Mode

In this mode:

- Frames are acquired continuously
- Data is displayed but not stored
- Used for alignment, focusing, and setup

Preview mode prioritizes responsiveness over data retention.

7.3.2 Single-Frame Capture Mode

Single-frame capture:

- Acquires a single image
- Saves the frame to disk
- Used for reference images or documentation

This mode uses the same exposure settings as preview unless overridden.

7.3.3 Multi-Frame Acquisition Mode

Multi-frame acquisition is used for:

- Speckle contrast computation
- Statistical analysis
- Temporal averaging

Frames are acquired in rapid succession and processed either in real time or post-acquisition.

7.3.4 Acquisition Timing and Synchronization

During controlled acquisition:

- Frame timing is managed by the camera clock
- Software timestamps frames
- Optional synchronization with external triggers (e.g., NI-DAQ)

Precise timing is essential for stimulation-based experiments.

7.4 Image Bit-Depth and Scaling

The camera produces high-bit-depth image data that must be correctly handled for accurate visualization and analysis.

7.4.1 Native Bit-Depth

The camera outputs images at **12-bit grayscale resolution**, meaning each pixel can represent 4096 intensity levels.

This high dynamic range is critical for:

- Speckle contrast sensitivity
 - Accurate intensity profiling
 - Noise analysis
-

7.4.2 Internal Data Representation

Internally:

- Images are stored as integer arrays
 - No compression is applied during acquisition
 - Full bit-depth is preserved for processing and saving
-

7.4.3 Visualization Scaling

For display purposes:

- 12-bit data is scaled to 8-bit for screen rendering
- Scaling does not modify underlying data
- Contrast adjustments affect only visualization

Users should avoid confusing visual contrast with actual intensity values.

7.4.4 Saturation Detection

The software monitors for:

- Pixel saturation at maximum intensity
- Clipped histograms
- Loss of dynamic range

Saturation warnings indicate exposure adjustment is required.

7.5 Camera Control Window Explained

The Camera Control Window is the **central interface for all camera-related parameters**.

7.5.1 Layout and Sections

The window is typically divided into the following sections:

1. **Exposure Control Section**
2. **ROI Configuration Section**
3. **Scale Calibration Section**
4. **Z-Axis Control Section**
5. **Status and Feedback Area**

Each section interacts with the live preview in real time.

7.5.2 Exposure Control Section

This section includes:

- Numeric exposure input field
- Increment/decrement buttons
- Apply/update controls

Exposure changes are reflected immediately during preview.

7.5.3 ROI Configuration Section

ROI (Region of Interest) settings allow:

- Selection of a subregion of the sensor
- Increased frame rate
- Reduced data processing load

Changing ROI:

- Affects image dimensions
 - May invalidate existing calibrations
 - Requires user confirmation in some cases
-

7.5.4 Scale Calibration Section

This section displays:

- Current pixel-to-millimeter scale
- Tools for updating scale calibration
- Visual rulers on the live image

Scale calibration is essential for quantitative spatial analysis.

7.5.5 Z-Axis Control Section

If enabled:

- Provides fine control over vertical positioning
 - Allows incremental motion commands
 - Integrates with live preview feedback
-

7.5.6 Status and Diagnostics

The control window displays:

- Camera state
- Current exposure
- ROI dimensions
- Frame rate (if available)

This information is essential for diagnosing acquisition issues.

7.6 Best Practices for Reliable Image Acquisition

- Always initialize the camera before adjusting parameters
 - Perform exposure optimization during preview
 - Avoid changing settings mid-capture
 - Monitor histograms regularly
 - Document camera settings for reproducibility
-

7.7 Summary

This chapter has provided a **complete technical and operational overview of camera control and image acquisition**. Mastery of these concepts ensures high-quality data, reliable speckle analysis, and reproducible experiments.

End of Chapter 7

Chapter 8 – Region of Interest (ROI) Selection and Management

8.1 Purpose of ROI

A Region of Interest (ROI) is a **user-defined subregion of the camera sensor** that limits image acquisition, display, processing, and storage to a specific area. ROI functionality is a critical feature for optimizing performance, improving analysis relevance, and maintaining experimental consistency.

8.1.1 Why ROI Is Used

ROI selection serves several key purposes:

- **Performance Optimization**
Reducing the image area decreases data throughput, enabling higher frame rates and lower processing latency.
 - **Focused Analysis**
Limits analysis to biologically or experimentally relevant regions.
 - **Noise Reduction**
Excludes irrelevant background regions that may introduce noise or bias.
 - **Data Size Management**
Reduces file size and storage requirements.
 - **Calibration Precision**
Enables more accurate spatial calibration within a controlled area.
-

8.1.2 ROI as a First-Class System Parameter

In this system, ROI is not merely a display feature. It directly influences:

- Camera acquisition dimensions
- Coordinate mapping
- Calibration validity
- Analysis algorithms
- Data saving formats

Because of this, ROI must be managed deliberately and consistently.

8.2 Enabling and Disabling ROI

ROI functionality can be enabled or disabled through the **Camera Control Window**.

8.2.1 Default ROI State

- By default, ROI is **disabled**
 - Full sensor resolution is used
 - All calibrations assume full-frame coordinates
-

8.2.2 Enabling ROI

To enable ROI:

1. Open the Camera Control Window
2. Locate the ROI configuration section
3. Activate the ROI toggle or checkbox
4. Define ROI boundaries (X, Y, width, height)
5. Apply the ROI settings

Once enabled:

- Camera acquisition switches to ROI mode
 - Display panels update to reflect the cropped region
 - ROI-dependent coordinate mapping becomes active
-

8.2.3 Disabling ROI

Disabling ROI:

- Restores full-frame acquisition
- Resets coordinate origin to the full sensor
- May invalidate ROI-based calibrations

Users should disable ROI before changing major system settings or recalibrating.

8.3 ROI Coordinate System

Understanding the ROI coordinate system is essential for calibration, galvo mapping, and quantitative analysis.

8.3.1 Full-Frame Coordinate System

When ROI is disabled:

- Origin (0,0) is at the top-left corner of the full sensor
 - Coordinates increase rightward (X) and downward (Y)
 - All pixel references are absolute
-

8.3.2 ROI-Relative Coordinate System

When ROI is enabled:

- Origin (0,0) is at the top-left corner of the ROI
- Coordinates are **relative to the ROI**, not the full sensor
- Full-frame pixel coordinates are internally remapped

For example:

- Full-frame pixel (800, 600)
 - ROI starting at (600, 400)
 - ROI-relative coordinate becomes (200, 200)
-

8.3.3 Internal Coordinate Translation

Internally, the software maintains:

- ROI offset (X_0 , Y_0)
- ROI width and height
- Transformation logic for converting between coordinate systems

This ensures consistency across:

- Display
 - Calibration
 - Analysis
 - Saving routines
-

8.4 ROI-Aware Display and Cursor Readout

The GUI dynamically adapts visual feedback based on ROI state.

8.4.1 Live Image Display Behavior

When ROI is enabled:

- Only the ROI region is displayed
- Image dimensions match ROI size
- Display scaling is recalculated automatically

This ensures that visual feedback matches acquired data exactly.

8.4.2 Cursor Coordinate Readout

Cursor readouts adapt to ROI state:

- **ROI disabled:**
Cursor shows full-frame pixel coordinates
- **ROI enabled:**
Cursor shows ROI-relative coordinates

This behavior is clearly indicated in the GUI to prevent confusion during calibration or analysis.

8.4.3 Rulers and Measurement Tools

If pixel-to-millimeter calibration is available:

- Rulers reflect ROI-relative spatial dimensions
- Measurements remain accurate within the ROI

Without calibration, rulers remain in pixel units.

8.5 ROI Effects on Calibration

ROI selection has **significant implications for calibration**, especially for galvo mapping and spatial scaling.

8.5.1 Galvo Calibration Dependency

Galvo calibration maps galvo angles to image pixel coordinates.

Because ROI changes pixel coordinates:

- Any change in ROI **invalidates existing galvo calibration**
- Calibration must be repeated after ROI modification

The system enforces this by disabling galvo functions when calibration is invalid.

8.5.2 Pixel-to-Scale Calibration

Pixel-to-millimeter calibration is also ROI-dependent:

- Calibration performed under one ROI may not apply to another
- Full-frame calibration does not automatically transfer to ROI mode

Users are encouraged to perform scale calibration under final ROI settings.

8.5.3 Software Safeguards

The software:

- Detects ROI changes
- Flags calibration as invalid when necessary
- Prompts the user to recalibrate

These safeguards prevent silent errors in spatial mapping.

8.6 ROI Effects on Analysis and Saving

ROI selection directly affects how data is analyzed and stored.

8.6.1 Analysis Scope

All analysis tools operate **exclusively on ROI data** when ROI is enabled:

- Speckle contrast computation
- Intensity profiles
- Histograms
- Temporal analysis

This ensures that analysis reflects exactly what is acquired.

8.6.2 Saved Image Dimensions

When saving images:

- Saved files contain only ROI pixels
- Metadata includes ROI size and offsets
- File names may encode ROI status

This behavior simplifies downstream analysis and reduces ambiguity.

8.6.3 Metadata and Traceability

Saved datasets include:

- ROI position and dimensions
- Exposure settings
- Calibration status

This ensures full traceability for reproducibility and audits.

8.7 Resetting and Validating ROI

Proper ROI management includes knowing when and how to reset or validate ROI settings.

8.7.1 Resetting ROI to Full Frame

To reset ROI:

1. Disable ROI in the Camera Control Window
2. Apply changes
3. Restart preview if required

This restores full-frame acquisition and resets coordinate mapping.

8.7.2 Validating ROI Settings

After defining ROI:

- Confirm live image dimensions
- Verify cursor coordinate behavior
- Check calibration status indicators

- Perform a short test acquisition

Validation should always precede calibration or data collection.

8.7.3 Common ROI Mistakes

- Changing ROI after calibration
 - Forgetting ROI is enabled
 - Mixing datasets with different ROI sizes
 - Performing scale calibration before ROI finalization
-

8.8 Summary

This chapter has explained the **conceptual, operational, and analytical role of ROI** in the imaging system. Proper ROI management is essential for accurate calibration, meaningful analysis, and reproducible results.

End of Chapter 8

Chapter 9 – Live Imaging Tools

9.1 Live Image Display Features

The Live Image Display is the primary visual feedback mechanism during system operation. It presents real-time camera data and serves as the foundation for alignment, focusing, ROI definition, calibration, and quality assessment.

9.1.1 Real-Time Image Rendering

When live preview is active, the system continuously acquires frames from the camera and renders them in the Live Image Display panel.

Key characteristics include:

- Continuous frame updates at the camera's preview frame rate
- Grayscale visualization of raw intensity data
- Immediate reflection of exposure, focus, and illumination changes
- Automatic resizing to fit the display panel while preserving aspect ratio

The live display is optimized for responsiveness rather than archival precision.

9.1.2 Image Orientation and Reference

The displayed image follows the camera's native orientation:

- Top of the image corresponds to the top of the sensor
- Left-to-right and top-to-bottom pixel indexing is consistent with coordinate readouts

Any orientation changes applied by the camera or optics should be documented externally, as the software assumes a fixed orientation.

9.1.3 Cursor Interaction and Feedback

The live image responds to cursor movement:

- Pixel coordinates are displayed in real time
- Coordinates update dynamically based on ROI state
- Cursor feedback assists in ROI definition, calibration point selection, and profile extraction

This interactive feedback is essential for precision tasks.

9.1.4 Overlay Elements

Depending on system state, the live display may include overlays such as:

- ROI boundaries
- Calibration markers
- Rulers or scale indicators
- Selection lines for intensity profiles

Overlays are purely visual and do not alter underlying image data.

9.2 Contrast and Visualization Controls

Visualization controls allow users to adjust how image data is displayed without modifying the actual acquired values.

9.2.1 Display Scaling and Intensity Mapping

Raw camera data is typically 12-bit, while display devices are 8-bit. The system performs dynamic scaling to map raw values into a visible range.

Users can:

- Adjust display contrast
- Enhance visibility of low-intensity features
- Suppress background noise visually

These adjustments are non-destructive and affect display only.

9.2.2 Histogram-Guided Visualization

The live imaging tools often include histogram visualization, either directly or via auxiliary windows.

Histograms allow users to:

- Assess dynamic range usage
- Detect saturation or underexposure
- Guide exposure and contrast adjustments

A well-exposed image typically uses a large portion of the available histogram without clipping.

9.2.3 Visual Saturation Indicators

When pixel values reach the maximum representable intensity:

- Saturation may be indicated visually
- Histogram clipping becomes apparent
- Contrast adjustments may no longer reveal detail

Saturation is a data quality issue and should be corrected at the exposure level, not through visualization controls.

9.2.4 Interaction with Speckle Visualization

While live imaging tools focus on raw intensity:

- Visualization choices influence perceived speckle clarity
- Overly aggressive contrast scaling can mislead interpretation

Users should confirm image quality using raw intensity distributions when preparing for speckle analysis.

9.3 Full-Frame vs ROI-Based Viewing

The live imaging system adapts dynamically depending on whether full-frame or ROI acquisition is active.

9.3.1 Full-Frame Viewing Mode

In full-frame mode:

- The entire camera sensor area is displayed
- Cursor coordinates refer to absolute sensor pixels
- Calibration overlays reference full-frame geometry

Full-frame viewing is preferred during:

- Initial alignment
- Global inspection

- ROI selection
-

9.3.2 ROI-Based Viewing Mode

When ROI is enabled:

- Only the ROI region is displayed
- Image dimensions correspond to ROI size
- Cursor coordinates are ROI-relative

This mode is optimized for:

- High-speed imaging
 - Focused analysis
 - Reduced processing load
-

9.3.3 Visual Indicators of ROI State

The GUI clearly indicates ROI state via:

- Visible ROI boundaries (when defining ROI)
- Changes in image dimensions
- Status indicators in control panels

Users should always verify ROI state before calibration or saving data.

9.3.4 Switching Between Modes

Switching between full-frame and ROI-based viewing:

- May briefly interrupt live preview
- Triggers internal reconfiguration
- May invalidate calibration

Such transitions should be performed deliberately and infrequently.

9.4 Live Image Saving

The live imaging tools include options to save the currently displayed image for documentation or analysis.

9.4.1 Purpose of Live Image Saving

Live image saving is typically used for:

- Recording reference images
- Documenting alignment or focus
- Capturing representative frames

Live-saved images are not intended to replace structured acquisition workflows.

9.4.2 Saving Behavior

When the save function is activated:

- The current frame is copied from the live buffer
- Image data is saved at full bit-depth
- The saved image reflects ROI state

Saved files include metadata such as:

- Exposure time
 - ROI parameters
 - Timestamp
-

9.4.3 File Naming and Organization

Saved images follow a consistent naming scheme and directory structure:

- Automatically generated filenames
- Separation between live images and experimental datasets
- Organized to support traceability

Users should avoid manual renaming that could obscure metadata relationships.

9.4.4 Limitations of Live Image Saving

- Only a single frame is saved
- No temporal context is preserved
- No speckle computation is performed

For quantitative studies, structured acquisition modes should be used instead.

9.5 Common Live Imaging Issues

Despite careful setup, issues may arise during live imaging. Understanding common symptoms and causes can significantly reduce downtime.

9.5.1 Black or Blank Image

Possible causes:

- Laser not emitting or blocked
- Exposure time too short
- Camera preview not started

Resolution:

- Verify illumination
 - Increase exposure
 - Confirm preview state
-

9.5.2 Overexposed or Saturated Image

Possible causes:

- Excessive exposure time
- Excessive laser intensity

Resolution:

- Reduce exposure
 - Adjust illumination
 - Confirm histogram behavior
-

9.5.3 Flickering or Unstable Image

Possible causes:

- Mechanical vibration
- Power supply instability
- USB bandwidth issues

Resolution:

- Secure optical components
 - Use direct USB connection
 - Reduce background system load
-

9.5.4 Lag or Low Frame Rate

Possible causes:

- Large image resolution
- Heavy background processing
- ROI disabled when not needed

Resolution:

- Enable ROI
 - Close unnecessary tools
 - Reduce preview resolution if supported
-

9.5.5 Mismatch Between Display and Saved Image

Possible causes:

- Display scaling confusion
- ROI misunderstanding

Resolution:

- Verify saved image dimensions
 - Inspect raw data rather than display rendering
-

9.6 Summary

This chapter has detailed the **tools, controls, and behaviors associated with live imaging**, emphasizing correct interpretation of real-time data and safe, effective operation. Live imaging is the foundation upon which all advanced imaging and analysis workflows are built.

End of Chapter 9

Chapter 10 – Speckle Contrast Imaging

10.1 Principles of Speckle Contrast

Speckle contrast imaging is a **non-invasive optical technique** that exploits the statistical properties of laser speckle patterns formed when coherent light scatters from a dynamic medium. In biological and flow-related samples, microscopic motion—such as red blood cell flow—induces temporal fluctuations in speckle intensity.

10.1.1 Formation of Laser Speckle

When coherent laser light illuminates a rough or scattering surface:

- Light waves interfere constructively and destructively
- A granular intensity pattern (speckle) is formed
- The pattern is highly sensitive to motion within the scattering volume

Static regions produce stable speckle, while moving scatterers cause rapid speckle fluctuations.

10.1.2 Definition of Speckle Contrast

Speckle contrast, denoted by **K**, is defined as:

$$K = \frac{\sigma_I}{\langle I \rangle}$$

where:

- σ_I is the standard deviation of intensity
- $\langle I \rangle$ is the mean intensity

Higher contrast corresponds to slower motion; lower contrast corresponds to faster motion.

10.1.3 Relationship to Flow and Dynamics

Speckle contrast is inversely related to motion:

- **High flow → rapid decorrelation → low contrast**

- **Low flow → slow decorrelation → high contrast**

While speckle contrast does not directly measure absolute velocity, it provides **robust relative flow information**.

10.2 Spatial vs Temporal Contrast

The system supports both **spatial** and **temporal** speckle contrast computation, each with distinct characteristics.

10.2.1 Spatial Speckle Contrast

Spatial contrast is computed over a small neighborhood of pixels within a single frame.

Characteristics:

- Uses a spatial window (e.g., 5×5 pixels)
- Requires only one frame
- Sensitive to spatial heterogeneity
- Lower temporal resolution

Advantages:

- Fast computation
- Suitable for static or slowly changing scenes

Limitations:

- Sensitive to static spatial artifacts
 - Reduced accuracy in heterogeneous illumination
-

10.2.2 Temporal Speckle Contrast

Temporal contrast is computed at each pixel across multiple frames.

Characteristics:

- Uses a sequence of frames
- Computes statistics over time
- Higher sensitivity to motion
- Requires stable illumination

Advantages:

- More accurate flow sensitivity
- Pixel-wise temporal resolution

Limitations:

- Requires multiple frames
 - Higher computational load
-

10.2.3 Hybrid Approaches

In practice, temporal contrast is often preferred for **dynamic biological imaging**, while spatial contrast may be used for rapid visualization or system testing.

10.3 Speckle Contrast Computation Pipeline

The speckle computation pipeline transforms raw camera data into quantitative contrast maps.

10.3.1 Data Acquisition

- A sequence of raw intensity frames is acquired
 - Frames are captured with fixed exposure
 - ROI cropping is applied if enabled
-

10.3.2 Preprocessing

Before contrast computation:

- Frames are converted to floating-point representation
 - Dark offsets (if any) are implicitly accounted for
 - Saturated pixels are identified
-

10.3.3 Statistical Computation

Depending on mode:

Temporal contrast:

- Mean intensity computed per pixel across frames
- Standard deviation computed per pixel

- Contrast calculated pixel-wise

Spatial contrast:

- Local neighborhood extracted
 - Mean and standard deviation computed spatially
-

10.3.4 Post-Processing

- Contrast values normalized
 - Optional inversion or scaling applied
 - Invalid pixels masked if necessary
-

10.3.5 Visualization Preparation

- Contrast values mapped to color scale
 - C-axis limits applied
 - Image rendered in speckle display panel
-

10.4 Frames-per-Speckle Configuration

Frames-per-speckle is a **critical parameter** in temporal speckle contrast computation.

10.4.1 Definition

Frames-per-speckle defines:

- The number of consecutive frames used to compute one speckle contrast image

Typical values:

- 5–10 frames: fast response, noisy
 - 20–50 frames: balanced
 - 100+ frames: smooth but slower temporal response
-

10.4.2 Trade-Offs

- **Fewer frames:**
 - Higher temporal resolution

- Increased noise
- **More frames:**
 - Reduced noise
 - Lower temporal resolution

The choice depends on experimental requirements.

10.4.3 GUI Configuration

Frames-per-speckle is configured via:

- Numeric input field
- Applied immediately to speckle computation
- Reflected in real-time speckle video

Users should avoid changing this parameter mid-experiment unless necessary.

10.5 Real-Time Speckle Video Mode

Speckle video mode enables **continuous real-time visualization** of speckle contrast maps.

10.5.1 Activation of Speckle Video

To start speckle video:

1. Ensure live preview is active
2. Set frames-per-speckle
3. Click “Start Speckle Video”

The system begins:

- Accumulating frames
 - Computing contrast continuously
 - Updating the speckle display panel
-

10.5.2 Internal Buffering Behavior

- A rolling buffer stores recent frames
- Old frames are discarded as new ones arrive
- Each update uses the most recent frame window

This ensures stable, real-time performance.

10.5.3 Synchronization with Live Imaging

Speckle video runs alongside live imaging:

- Live intensity feed remains available
 - ROI and exposure changes affect speckle output
 - Preview must remain active
-

10.6 C-Axis ($1/K^2$) Scaling and Color Mapping

Visualization scaling is essential for meaningful interpretation of speckle contrast.

10.6.1 Contrast-to-Flow Mapping

Often, speckle contrast is visualized using:

$$\text{Flow Index} \propto \frac{1}{K^2}$$

This transformation:

- Enhances sensitivity to flow differences
 - Produces intuitive contrast maps
 - Improves visualization of low-flow regions
-

10.6.2 C-Axis Control

The C-axis slider:

- Controls minimum and maximum display values
- Does not affect raw contrast data
- Adjusts only the colormap scaling

Proper C-axis adjustment is crucial for:

- Avoiding saturation
- Highlighting physiologically relevant differences

10.6.3 Color Mapping

False-color maps are applied to speckle contrast images:

- Low flow → cooler colors
- High flow → warmer colors

The colorbar provides quantitative reference for interpretation.

10.7 Speckle Image Saving and Export

Speckle contrast images can be saved for documentation and analysis.

10.7.1 Saving Behavior

When saving a speckle image:

- The computed contrast image is saved
 - Raw contrast values are preserved
 - ROI state is reflected in image dimensions
-

10.7.2 Metadata Included

Saved files include:

- Exposure time
- Frames-per-speckle
- ROI parameters
- Scaling mode (K or $1/K^2$)
- Timestamp

This metadata ensures reproducibility.

10.7.3 Export Use Cases

Speckle images may be used for:

- Offline quantitative analysis
- Longitudinal comparison

- Publication figures
- Machine learning workflows

Users should avoid mixing datasets with different speckle parameters without documentation.

10.8 Summary

This chapter has provided a **complete conceptual and operational overview of speckle contrast imaging**, from physical principles to real-time GUI implementation. Proper understanding of speckle contrast computation and visualization is essential for reliable flow-sensitive imaging and quantitative interpretation.

End of Chapter 10

Chapter 11 – Intensity Analysis Tools

11.1 Overview of Intensity Profiling

Intensity analysis tools provide **quantitative insight into spatial and statistical variations** within both raw intensity images and processed speckle contrast images. Unlike qualitative visualization, intensity profiling enables users to extract numerical information that can be used for comparison, validation, and downstream analysis.

11.1.1 Role of Intensity Analysis

Intensity analysis is commonly used to:

- Examine illumination uniformity
- Detect spatial gradients or boundaries
- Compare regions across time or conditions
- Validate focus and exposure
- Support calibration and quality assurance

These tools are applicable to both **live imaging** and **speckle-based outputs**, but interpretation depends on the image type.

11.1.2 Image Types Supported

The system supports intensity analysis for:

- Live raw intensity images
- Speckle contrast images (K or $1/K^2$ representations)

Each image type has distinct intensity meaning and units, which must be considered during interpretation.

11.2 Live Image Intensity Profile

The live image intensity profile tool extracts pixel intensity values along a user-defined line from the raw camera image.

11.2.1 Purpose of Live Intensity Profiles

Live intensity profiles are primarily used to:

- Assess illumination gradients
- Detect saturation or underexposure
- Verify focus sharpness
- Identify structural boundaries

They provide immediate feedback during setup and alignment.

11.2.2 Accessing Live Intensity Profiles

To access the live intensity profile:

1. Ensure live preview is active
2. Click the “Live Intensity Profile” button
3. Draw a line on the live image display

The system immediately computes and displays the intensity profile.

11.2.3 Interpretation of Live Profiles

- X-axis: Distance along the line (pixels or calibrated units)
- Y-axis: Raw intensity values (camera counts)

Sharp transitions may indicate edges or features, while smooth gradients often reflect illumination changes.

11.3 Speckle Image Intensity Profile

Speckle intensity profiling applies the same concept to speckle contrast images.

11.3.1 Purpose of Speckle Profiles

Speckle profiles are used to:

- Compare relative flow across spatial regions
- Examine gradients in speckle contrast
- Validate physiological responses

Unlike raw intensity, speckle values reflect **dynamic properties**, not brightness.

11.3.2 Accessing Speckle Profiles

To generate a speckle intensity profile:

1. Ensure speckle video or speckle image is available
2. Click the “Speckle Intensity Profile” button
3. Draw a line on the speckle display

The profile updates based on current speckle scaling.

11.3.3 Interpreting Speckle Profiles

- Lower speckle contrast (or higher $1/K^2$) corresponds to higher flow
 - Profiles may show smoother variations compared to raw intensity
 - Interpretation must consider exposure and frames-per-speckle settings
-

11.4 Line Drawing and Interaction

Line-based interaction is central to intensity profiling.

11.4.1 Drawing a Profile Line

Users define a line by:

- Clicking to define a start point
- Dragging to define the end point
- Releasing to finalize the line

The system samples pixel values along the line using interpolation.

11.4.2 Editing and Repositioning Lines

Depending on implementation:

- Lines may be redrawn for each analysis
- Previous lines may be cleared automatically
- Line orientation and length affect resolution

Users should redraw lines carefully for consistent comparisons.

11.4.3 ROI and Line Interaction

When ROI is enabled:

- Line coordinates are ROI-relative
 - Profiles reflect only ROI data
 - Changing ROI invalidates previous line references
-

11.5 Statistical Outputs

In addition to visual profiles, the system computes basic statistical metrics.

11.5.1 Common Statistical Metrics

Typical outputs include:

- Mean intensity
- Standard deviation
- Minimum and maximum values
- Range

These metrics are computed along the drawn line or across selected regions.

11.5.2 Use of Statistics

Statistical outputs are useful for:

- Quantitative comparison across trials
- Assessing variability
- Validating imaging stability

Users should record these values alongside experimental metadata.

11.6 Histogram Analysis

Histogram analysis provides a **global statistical view** of intensity distribution.

11.6.1 Live Image Histogram

The live image histogram shows:

- Distribution of raw pixel intensities
- Dynamic range usage
- Saturation or underexposure

This tool is essential for exposure optimization.

11.6.2 Speckle Image Histogram

Speckle histograms display:

- Distribution of contrast or flow index values
- Shifts due to physiological changes
- Overall heterogeneity

Histogram shape provides insight beyond mean values.

11.6.3 Interpreting Histogram Features

- Narrow peak → uniform region
- Broad distribution → heterogeneous region
- Clipping → data quality issue

Histogram interpretation should always consider visualization scaling.

11.7 Exporting Profiles and Reports

The system supports exporting intensity analysis results for documentation and offline analysis.

11.7.1 Exported Data Types

Export options may include:

- Profile data as text or CSV
- Histogram data

- Snapshot images of plots
 - Summary statistics
-

11.7.2 Metadata and Traceability

Exported files include:

- Timestamp
- Image type (live or speckle)
- ROI state
- Exposure and speckle parameters

This ensures reproducibility and auditability.

11.7.3 Best Practices for Reporting

- Export profiles using consistent line placement
 - Label datasets clearly
 - Avoid mixing different scaling modes
 - Archive raw data alongside processed results
-

11.8 Summary

This chapter has described the **full suite of intensity analysis tools**, covering live and speckle images, interactive profiling, statistical outputs, histogram analysis, and data export. Proper use of these tools enables quantitative insight and supports rigorous experimental analysis.

End of Chapter 11

Chapter 12 – Calibration Concepts and Procedures: Scale and Galvo Mapping

12.1 Purpose and Importance of Calibration

Calibration is the process by which the imaging system establishes a **quantitative relationship between physical space, camera pixels, and galvo mirror control parameters**. Without proper calibration, measurements derived from images—such as distances, spatial profiles, or galvo-directed targeting—cannot be interpreted reliably.

In this system, calibration serves three primary purposes:

- **Spatial Quantification**
Converts pixel measurements into real-world physical units (e.g., millimeters).
- **Beam Targeting Accuracy**
Ensures that galvo mirror angles correspond correctly to image coordinates.
- **Reproducibility and Traceability**
Guarantees that experiments can be repeated and compared across sessions.

Calibration is therefore not optional; it is a **foundational requirement** for all quantitative imaging and galvo-based experiments.

12.2 Types of Calibration in the System

The system supports two major categories of calibration:

1. **Scale (Pixel-to-Physical Unit) Calibration**
2. **Galvo Mapping (Angle-to-Pixel Calibration)**

These calibrations are **independent but interrelated**, and both are affected by ROI selection, optical configuration, and camera settings.

12.3 General Calibration Principles

Before performing any calibration, the following principles must be understood and respected.

12.3.1 Calibration Is Configuration-Specific

Calibration results are valid **only under the conditions in which they were obtained**, including:

- Camera resolution
- ROI settings
- Optical magnification
- Working distance
- Galvo mounting geometry

Any change to these parameters requires recalibration.

12.3.2 Calibration Is State-Dependent

Calibration assumes that:

- The camera is initialized
- Live preview is active
- Exposure and focus are finalized
- ROI is fixed

Attempting calibration under unstable conditions will result in inaccurate mappings.

12.3.3 Calibration Validity Tracking

The software tracks calibration validity internally.

When a calibration is invalidated (e.g., due to ROI change), galvo-dependent features are automatically disabled to prevent misuse.

12.4 Scale Calibration (Pixel-to-Physical Mapping)

Scale calibration establishes the relationship between **image pixels and real-world physical distance**.

12.4.1 Purpose of Scale Calibration

Scale calibration allows users to:

- Measure distances in physical units
- Quantify spatial features
- Report results in standardized units
- Enable meaningful comparison across experiments

Without scale calibration, all measurements remain in pixel units.

12.4.2 Required Equipment

Typical scale calibration requires:

- A calibration target (e.g., ruler, grid, micrometer slide)
 - Known physical spacing
 - Stable illumination
 - Clear focus across the target
-

12.4.3 Scale Calibration Procedure

A typical procedure includes:

1. Place the calibration target within the field of view
2. Start live preview
3. Adjust exposure and focus for clear visibility
4. Draw a line between two known points on the target
5. Enter the known physical distance
6. Compute the pixel-to-unit conversion factor
7. Save the calibration

The system stores the scale factor and applies it to all subsequent measurements.

12.4.4 Application of Scale Calibration

Once applied:

- Rulers display real-world units
 - Intensity profiles use physical distance axes
 - ROI dimensions are reported in physical units
 - Exported data includes calibrated dimensions
-

12.4.5 Limitations and Common Errors

Common issues include:

- Poor focus during calibration
- Incorrect identification of reference points
- Calibration performed before ROI finalization

- Using a target not aligned with the imaging plane

These errors can lead to systematic measurement bias.

12.5 Galvo Mapping Calibration Overview

Galvo mapping calibration establishes the relationship between **galvo mirror control parameters (angles or voltages)** and **image pixel coordinates**.

12.5.1 Why Galvo Calibration Is Necessary

Galvo mirrors steer the laser beam, but:

- Their mechanical geometry is nonlinear
- Optical paths introduce distortions
- Mounting and alignment vary across systems

Calibration compensates for these factors to enable precise beam targeting.

12.5.2 What Galvo Calibration Produces

The calibration process generates:

- A mapping function or matrix
- Coefficients relating galvo inputs to pixel locations
- Residual error metrics

This mapping is stored and reused until invalidated.

12.6 Galvo Calibration Preparation

Before beginning galvo calibration, ensure:

- Scale calibration is complete (recommended)
- ROI is finalized
- Camera exposure and focus are optimized
- Laser spot is visible and well-defined
- Galvo movement range is known and safe

Calibration should be performed under **identical conditions** to those used during experiments.

12.7 Galvo Mapping Procedure

The galvo calibration process typically follows a multi-point mapping approach.

12.7.1 Point-Based Calibration Concept

Calibration is performed by:

- Commanding the galvo to predefined positions
- Observing the laser spot location in the image
- Recording corresponding pixel coordinates

Multiple points are used to characterize system behavior.

12.7.2 Step-by-Step Galvo Calibration

A typical workflow includes:

1. Enable galvo control
2. Select calibration mode
3. Move galvo to a reference position
4. Identify and confirm laser spot location
5. Record pixel coordinates
6. Repeat for multiple positions across the field
7. Compute the mapping model
8. Evaluate fitting error
9. Save calibration results

The software may use linear or polynomial fitting depending on implementation.

12.7.3 Calibration Grid and Coverage

Good calibration requires:

- Coverage across the usable field of view
- Avoidance of edge clipping
- Uniform sampling in X and Y

Sparse or clustered calibration points reduce accuracy.

12.8 Galvo Calibration Models and Accuracy

12.8.1 Mapping Models

Typical mapping models include:

- Linear transformation
- Affine mapping
- Polynomial regression

The chosen model balances accuracy and computational stability.

12.8.2 Error Metrics

After fitting, the system reports:

- Residual errors
- Maximum deviation
- Average mapping error

Users should verify that errors are within acceptable experimental limits.

12.8.3 Validation of Calibration

Validation steps include:

- Moving the galvo to random positions
- Verifying spot alignment visually
- Comparing expected and observed positions

Calibration should be repeated if errors are unacceptable.

12.9 ROI and Calibration Interdependence

ROI directly affects calibration validity.

12.9.1 ROI Change Effects

Changing ROI:

- Shifts pixel coordinate origin
- Alters mapping scale
- Invalidates galvo calibration

The system enforces recalibration after ROI modification.

12.9.2 Best Practice Sequence

Recommended order:

1. Define ROI
 2. Perform scale calibration
 3. Perform galvo calibration
 4. Lock settings for experiments
-

12.10 Saving, Loading, and Managing Calibration Files

12.10.1 Calibration File Storage

Calibration data is stored in:

- Dedicated calibration directories
- Timestamped files
- Human-readable and machine-readable formats

Backup copies are created automatically.

12.10.2 Loading Calibration on Startup

On application launch:

- The most recent valid calibration is loaded
 - Validity is checked against current settings
 - Users are warned if mismatches occur
-

12.10.3 Versioning and Traceability

Each calibration file includes:

- Date and time
- ROI parameters
- Camera settings
- Calibration type

This supports long-term traceability.

12.11 Summary

This chapter has provided a **complete conceptual and procedural framework** for scale and galvo calibration. Accurate calibration underpins every quantitative measurement and galvo-based operation in the system, making it one of the most critical components of reliable imaging workflows.

End of Chapter 12

Chapter 13 – Galvo System, Scanning, and Spatial Targeting

13.1 Galvo Hardware Overview

The galvo system is a **dual-axis mirror setup** that allows precise steering of the laser beam across the imaging field. It consists of:

- **X-axis and Y-axis galvo mirrors**
Reflect the laser to target specific positions in the sample plane.
- **Galvo motor drivers**
Control mirror angles based on input voltages or digital commands.
- **Feedback sensors (optional)**
Provide closed-loop position monitoring for improved accuracy.
- **Interface to GUI**
Serial or USB communication allows integration with imaging and analysis software.

Galvo mirrors are highly responsive and capable of rapid angular adjustments, enabling grid scanning, targeted illumination, and precise synchronization with camera acquisition.

13.2 Galvo Enable and Safety Interlocks

13.2.1 Purpose of Enable Switch

The galvo enable control ensures that mirrors are not driven unintentionally. Activation involves:

- Powering the galvo drivers
 - Allowing control signals to reach the motors
 - Engaging software interlocks
-

13.2.2 Safety Interlocks

Interlocks prevent accidental laser movement, which could cause:

- Damage to optical components
- Misaligned scans
- Safety hazards

Common interlocks include:

- **Hardware-level limits** preventing mirror over-rotation
- **Software checks** ensuring calibration validity before activation
- **Emergency stop integration** halting motion instantly

Before enabling, ensure that:

- Calibration is valid
 - ROI is defined
 - Laser power is safe
-

13.3 Galvo Dial Controls

The GUI provides **virtual dials** for manual control of galvo angles.

13.3.1 Functionality

- X and Y angles can be adjusted independently
 - Dial values correspond to angular displacement (degrees or voltage equivalent)
 - Real-time updates display the current mirror position
-

13.3.2 Operational Considerations

- Dials are disabled if calibration is invalid or if the system detects unsafe conditions
 - Adjustments are limited by **motion limits** to prevent overtravel
 - Movements are reflected in live preview for alignment verification
-

13.4 Manual Angle Entry

Manual angle entry allows **precise numeric input** for galvo positioning.

13.4.1 Manual Angle Entry Interface

- Accessible through the Manual Angle Entry window
 - Users can input exact angles for X and Y axes
 - Ideal for reproducible positioning or targeted illumination
-

13.4.2 Input Format and Conventions

- Angles may be entered in **degrees** or **normalized units** depending on system settings
 - Negative and positive values correspond to mirror orientation relative to central reference
 - Validation checks prevent out-of-range values
-

13.4.3 Using Current Dial Values

- Existing dial positions can populate the manual entry fields
 - Allows incremental adjustments or fine-tuning without resetting other values
-

13.4.4 Previewing Combined Calibration

- Entered angles are visualized in the live image or overlay before confirming movement
 - This ensures intended targeting aligns with calibration mapping
-

13.4.5 Clearing and Editing Entries

- Manual entries can be reset to zero or previous values
 - Editing allows iterative calibration or scanning sequences without restarting the GUI
-

13.5 Coordinate Feedback and Status

The system continuously provides feedback for each galvo axis:

- Current angle displayed in the GUI
- Corresponding pixel coordinates mapped via calibration
- Status indicators (enabled/disabled, error state, limit reached)

This feedback is essential for real-time verification of targeting accuracy.

13.6 Galvo Motion Limits

13.6.1 Hardware and Software Limits

- Physical limits prevent the mirror from exceeding safe angular range

- Software imposes additional constraints based on ROI and calibration boundaries
-

13.6.2 Operational Safety

- Always verify that intended positions lie within limits
 - Attempted motion beyond limits triggers warnings and is blocked
-

13.7 Automatic Dial-and-Click Calibration

Automatic calibration maps galvo angles to image coordinates using **user-selected reference points**.

13.7.1 Procedure

1. Enable calibration mode
2. Move the galvo to a target region
3. Click the corresponding point in the live image
4. Repeat for multiple points forming a calibration grid

The system automatically fits a transformation model based on collected pairs.

13.7.2 Advantages

- Reduces manual entry errors
 - Captures system nonlinearities
 - Quick and repeatable
-

13.8 Manual Angle Calibration

Manual calibration involves entering angles numerically or via dial to match target points in the image:

- Allows precise control in case of irregular calibration patterns
 - Useful when automated detection is not feasible
 - Typically used in **low-density calibration grids** or when working with partial fields
-

13.9 Hybrid Calibration (Manual + Dial)

Hybrid calibration combines automatic and manual steps:

- Automatic dial-and-click calibration for coarse mapping
 - Manual angle input to refine positions
 - Useful when field-of-view distortions or nonlinearities are significant
-

13.9.1 Calibration Grid Options

- Standard grids: 2×2 , 3×3 , or user-defined
 - Grid density affects accuracy versus calibration time
 - Uniformly spaced grids are recommended for consistent regression results
-

13.9.2 Step-by-Step Hybrid Calibration Workflow

1. Define ROI and exposure
 2. Enable live preview
 3. Perform automatic dial-and-click calibration on coarse grid
 4. Review mapped points
 5. Enter manual angles for fine-tuning
 6. Confirm alignment visually in live image
 7. Compute final mapping matrix
 8. Save calibration and validate
-

13.9.3 Calibration Completion and Validation

After hybrid calibration:

- Display the residual error for each grid point
 - Verify that target coordinates match expected positions
 - Perform random test points to ensure coverage
 - Save calibration file with timestamp and metadata
-

13.10 Best Practices for Galvo Calibration and Scanning

- Always calibrate **after ROI and exposure are finalized**
- Avoid mechanical adjustments during calibration
- Validate mapping using random target positions
- Keep detailed records of calibration parameters for reproducibility

- Combine live feedback with manual adjustments to achieve optimal alignment

13.11 Summary

This chapter has provided a **complete guide to the galvo system**, including hardware overview, enabling, safety, manual and dial controls, motion limits, and all calibration methods. Mastery of these procedures ensures **precise spatial targeting**, reproducible scanning, and reliable integration with both live imaging and speckle contrast workflows.

End of Chapter 13

Chapter 14 – Galvo Grid Scanning

Galvo Grid Scanning enables automated, repeatable spatial traversal of predefined galvo target positions derived from calibration. This chapter explains how the system performs grid-based scanning, how scan execution is controlled, how timing and ordering are handled, and how positional data and results are logged for documentation and reproducibility.

14.1 Scan Modes and Use Cases

Overview

Galvo Grid Scanning operates by sequentially moving the galvo mirrors to a set of calibrated angular coordinates arranged in a 2D grid (rows $\times$ columns). Each grid point corresponds to a specific spatial location in the imaging field, previously established through calibration.

The scan system is designed to work reliably across multiple experimental paradigms, including imaging, stimulation, validation, and characterization tasks.

Supported Scan Mode

The current implementation uses a **discrete point-by-point grid scan**, where the galvos move to each grid coordinate, stabilize, and optionally capture imaging data before proceeding to the next point.

Key characteristics:

- Deterministic, ordered traversal of grid points
- Fixed dwell and safety delays between movements
- Tight integration with image capture and logging

Common Use Cases

- **Spatial response mapping**
Sequentially probe responses across the field of view using consistent galvo positions.
- **Calibration verification**
Revisit all calibration grid points to visually or quantitatively validate mapping accuracy.
- **Laser or illumination uniformity testing**
Measure intensity or speckle statistics at known spatial locations.
- **Automated acquisition experiments**
Combine galvo motion with camera capture for structured data collection.
- **Repeatability and drift analysis**
Re-scan identical grid patterns over time to assess mechanical or optical drift.

14.2 Scan Order and Timing

Scan Order

Grid scanning follows a **row-major order**:

1. Start at **Row 1, Column 1**
2. Increment column index across the row
3. Move to the next row after completing all columns
4. Continue until the final grid point (Row N, Column M)

Mapping logic:

```
index → (row, col)
row = index // number_of_columns
col = index % number_of_columns
```

This predictable ordering simplifies:

- Correlation with saved image filenames
- Structured logging and documentation
- Post-processing and reconstruction

Galvo Motion Sequence

For each grid point:

1. **X-axis galvo moves to target angle**
2. Fixed stabilization delay
3. **Y-axis galvo moves to target angle**
4. Fixed stabilization delay
5. Optional image capture and logging

This sequential axis motion ensures mechanical stability and reduces transient vibration or overshoot.

Timing and Delays

Several timing elements are intentionally enforced:

- **Inter-axis delay** ($X \rightarrow Y$): allows mirror stabilization
- **Post-movement delay**: ensures final settling before capture
- **Inter-point delay**: safety buffer before moving to the next grid position

These delays are critical for:

- Protecting galvo hardware

- Ensuring repeatable optical alignment
 - Avoiding motion blur or transient artifacts in captured data
-

14.3 Start, Pause, Resume, and Stop

Starting a Grid Scan

When a scan is started:

- The system verifies that a valid calibration matrix is loaded
- ROI parameters (if enabled) are synchronized
- Progress indicators are initialized
- A dedicated worker thread is launched for scanning

Once started, the scan proceeds automatically through the grid unless interrupted.

Pausing a Scan

A scan can be paused safely at any time:

- The current grid index is preserved
- Motion stops after the current point completes
- No data corruption occurs

This is useful for:

- Mid-experiment inspection
- Temporary hardware adjustments
- Operator intervention without restarting the scan

Resuming a Scan

When resumed:

- Scanning continues from the last unfinished grid point
- Grid order and indexing remain unchanged
- Logged data remains continuous and consistent

This design ensures that partial scans can be completed without redundancy or ambiguity.

Stopping a Scan

Stopping the scan:

- Terminates further grid traversal
- Ends the scan thread cleanly
- Leaves the galvo at the last visited position

Stopping is intended for early termination rather than temporary interruption.

14.4 Coordinate Logging and Documentation

Automatic Logging

For every grid position visited, the system records:

- Grid row and column index
- Galvo X and Y angles (in degrees or system units)
- Timestamped image filename (if captured)

Log File Format

Logs are stored in a continuously updated **Word document** (**.docx**) named:

```
coordinate_log.docx
```

Each entry includes a human-readable description such as:

- Grid position
- Exact galvo angles
- Associated image file path

This format was chosen to:

- Allow easy review by non-technical users
- Support manual annotation
- Enable direct inclusion in experiment records or reports

Image Naming Convention

Captured images follow a structured naming pattern:

```
point_r<row>_c<col>_<timestamp>.png
```

Benefits:

- Immediate traceability to grid location
- Chronological sorting
- Simplified batch analysis

14.5 Returning to Reference Position

Reference Position Concept

A reference position typically corresponds to:

- The optical center
- A known neutral galvo angle
- A safe parking position for the mirrors

While the grid scan itself ends at the final grid point, returning to a reference position is an important operational consideration.

Manual Return

After scan completion or interruption, the user may:

- Manually set galvo dials to the reference angles
- Use direct angle entry to command the reference position

Best Practices

It is recommended to:

- Return to a reference position before disabling galvo power
- Re-center before switching experiments
- Use the same reference position consistently across sessions

This minimizes cumulative alignment errors and simplifies calibration reuse.

14.6 Summary

Galvo Grid Scanning provides a structured, automated mechanism for spatially resolved experiments. By combining calibrated grid coordinates, controlled timing, safe execution controls, and comprehensive logging, the system ensures high repeatability, traceability, and experimental reliability.

This chapter completes the operational workflow from calibration to automated spatial scanning, forming the foundation for advanced imaging and stimulation experiments.

End of Chapter 14

Chapter 15 – Manual Angle Input Tool and Hybrid Galvo Calibration

15.1 Overview and Purpose

The Manual Angle Input Tool provides **direct, numerical control** over galvo mirror positions and plays a critical role in **advanced calibration workflows**, particularly hybrid calibration strategies that combine manual precision with interactive GUI-based mapping.

This tool is essential when:

- Automated calibration is insufficient or ambiguous
- Fine correction is required near critical regions
- Reproducible, exact galvo positioning is needed
- Calibration must be refined after coarse alignment

Manual angle input enables users to operate at a **deterministic, parameter-level**, rather than relying solely on visual or click-based interaction.

15.2 Manual Angle Entry Interface

15.2.1 Interface Layout

The Manual Angle Input Tool is accessed through a dedicated control window within the galvo calibration section of the GUI. The interface typically includes:

- Numeric input fields for **X-axis angle**
- Numeric input fields for **Y-axis angle**
- Buttons to apply, preview, or clear entries
- Indicators for current galvo position
- Validation and status messages

The interface is designed to allow **precise, intentional movements** without ambiguity.

15.2.2 Input Units and Representation

Depending on system configuration, manual angle values may represent:

- Physical angles (degrees)

- Normalized units
- Voltage-equivalent values sent to galvo drivers

The GUI enforces a consistent convention across:

- Dial controls
- Manual entry fields
- Calibration mapping logic

Users should confirm unit interpretation before entering values.

15.3 Input Format, Validation, and Conventions

15.3.1 Accepted Input Format

Manual entries:

- Accept signed numerical values
- Support decimal precision
- Are axis-independent (X and Y entered separately)

Invalid formats (e.g., non-numeric entries) are rejected automatically.

15.3.2 Range Validation

To prevent unsafe operation:

- Inputs are checked against predefined galvo motion limits
- Out-of-range values trigger warnings
- Commands beyond limits are blocked

This ensures that manual input cannot damage hardware or exceed optical boundaries.

15.3.3 Sign Conventions

Sign conventions define directionality:

- Positive and negative values correspond to mirror rotation relative to the center reference
- Conventions are consistent across dials, manual entry, and calibration models

Users should avoid mixing conventions between calibration sessions.

15.4 Using Current Dial Values

The manual angle tool integrates tightly with galvo dial controls.

15.4.1 Transferring Dial Values

Users can:

- Populate manual entry fields with current dial values
- Use the dial for coarse positioning
- Switch to manual entry for fine adjustments

This hybrid interaction allows efficient navigation across the galvo workspace.

15.4.2 Incremental Refinement

A common workflow involves:

1. Moving to approximate position using dials
2. Transferring dial values to manual input
3. Making small numerical adjustments
4. Applying refined position

This approach minimizes overshoot and improves repeatability.

15.5 Coordinate Preview and Feedback

15.5.1 Previewing Manual Entries

Before applying movement:

- The system may preview expected image-space coordinates
- Calibration mapping is used to estimate target position
- Visual overlays may indicate predicted spot location

This preview helps prevent unintended targeting.

15.5.2 Live Coordinate Feedback

After applying manual input:

- The galvo moves to the specified angles
- Live image updates to show laser spot
- Mapped pixel coordinates are displayed in the GUI

This feedback loop ensures transparency and user confidence.

15.6 Role of Manual Angle Input in Calibration

Manual angle input is not merely a positioning tool; it is a **core calibration instrument**.

15.6.1 Manual Angle Calibration

In fully manual calibration:

- The user enters known galvo angles
- Observes corresponding laser spot position
- Records pixel coordinates
- Repeats across multiple positions

This method is useful when:

- Automated click detection is unreliable
 - External reference points are used
 - Calibration geometry is constrained
-

15.6.2 Limitations of Pure Manual Calibration

- Time-consuming
- Sensitive to human error
- Requires careful documentation

For this reason, manual calibration is often combined with automated methods.

15.7 Hybrid Galvo Calibration Concept

Hybrid galvo calibration combines:

- **Automated dial-and-click calibration** for coarse mapping
- **Manual angle input** for fine correction

This approach balances speed and precision.

15.7.1 Motivation for Hybrid Calibration

Hybrid calibration is especially useful when:

- Optical distortions vary across the field
 - Edge regions require extra precision
 - Mechanical alignment is imperfect
 - High-accuracy targeting is required
-

15.7.2 Calibration Grid Density

Hybrid calibration supports grid sizes such as:

- 2×2 (minimal, fast)
- 3×3 (balanced)
- Higher-density grids for precision-critical applications

Manual angle input is often applied selectively to refine grid points.

15.8 Step-by-Step Hybrid Calibration Workflow

A typical hybrid calibration workflow proceeds as follows:

1. Finalize ROI and exposure
2. Enable live preview and ensure laser visibility
3. Perform automated dial-and-click calibration on grid points
4. Review mapping errors or visual mismatches
5. Select grid points requiring refinement
6. Use manual angle input to adjust galvo angles
7. Confirm laser spot alignment visually
8. Update calibration dataset
9. Recompute mapping model
10. Validate calibration using random test points
11. Save calibration

This structured approach ensures both efficiency and accuracy.

15.9 Editing, Clearing, and Reusing Manual Entries

15.9.1 Clearing Entries

Manual input fields can be:

- Cleared to zero
- Reset to previous values
- Overwritten with new values

This allows rapid iteration during calibration.

15.9.2 Editing Calibration Points

Previously recorded calibration points can be:

- Edited numerically
- Replaced with updated measurements
- Removed if deemed unreliable

Careful editing improves model robustness.

15.10 Validation and Error Assessment

After hybrid calibration:

- Residual mapping errors are displayed
- Users should test multiple arbitrary positions
- Spot alignment should be confirmed visually

If errors exceed acceptable limits, additional manual refinement is recommended.

15.11 Best Practices for Manual and Hybrid Calibration

- Use automated calibration for coarse mapping
- Apply manual input sparingly and intentionally
- Document all manual adjustments
- Validate across the full field of view

- Save calibration immediately after completion

15.12 Summary

This chapter has detailed the **Manual Angle Input Tool** and its critical role in **hybrid galvo calibration workflows**. By combining numerical precision with interactive feedback, the system enables highly accurate, reproducible galvo mapping even under challenging optical or mechanical conditions. Mastery of these tools is essential for advanced spatial targeting and scanning applications.

End of Chapter 15

Chapter 16 – Calibration Management

Calibration Management defines how galvo calibration data is persistently stored, automatically retrieved, safeguarded through backups, and restored when needed. Because calibration establishes the mapping between image-space coordinates and physical galvo angles, it is a foundational component of the system. Proper management ensures spatial accuracy, reproducibility across sessions, and long-term reliability of experiments.

This chapter describes how calibration data is handled automatically by the software, how users can inspect calibration state, how backups are created and restored, how calibration files are structured, and which best practices should be followed to maintain calibration integrity.

16.1 Auto-Save and Auto-Load Behavior

Automatic Saving of Calibration

The system implements a **fail-safe auto-save mechanism** for calibration data. Whenever a valid calibration matrix is generated or updated, it is immediately written to disk without requiring any manual user action.

Auto-save is triggered when:

- A new galvo calibration grid is generated
- Manual or hybrid calibration inputs are finalized
- Regression-based calibration matrices are computed
- An existing calibration matrix is replaced or updated
- ROI-aware calibration parameters are modified

The following data is saved:

- Galvo calibration matrix (grid of X–Y galvo angles)
- Grid dimensions (number of rows and columns)
- Calibration method identifier (automatic, manual, hybrid, regression)
- ROI offsets and ROI dimensions (if ROI is enabled)
- Camera resolution context
- Timestamp of calibration creation or update

This design minimizes the risk of data loss due to unexpected shutdowns, crashes, or user oversight.

Automatic Loading at Startup

On application startup:

1. The system checks the calibration directory for existing calibration files.
2. Binary calibration files are prioritized for loading.
3. If binary files are unavailable, JSON calibration files are used as a fallback.

When a valid calibration is found:

- The calibration matrix is loaded into memory
- Grid dimensions and calibration type are restored
- ROI parameters are reinstated if applicable
- The system becomes immediately ready for galvo scanning and coordinate targeting

If no calibration is found:

- The system enters an uncalibrated state
- Galvo grid scanning and coordinate conversion remain disabled
- The user is prompted to perform calibration before proceeding

This auto-load behavior ensures continuity across sessions and reduces setup time.

16.2 Viewing Calibration Status

Calibration Status Interface

The GUI provides a dedicated **Calibration Management window** that allows users to inspect the current calibration state in a clear, structured, and read-only format.

Displayed information typically includes:

- Whether a calibration is currently loaded
- Calibration grid size (e.g., 2×2 , 3×3 , or rectangular grids)
- Calibration type (regression, manual, hybrid, backup-restored)
- Shape and dimensionality of the calibration matrix
- Presence of calibration files on disk
- Number of available backup files

ROI Awareness

If Region of Interest (ROI) mode is active:

- ROI activation status is shown
- ROI coordinate boundaries are displayed
- This ensures users can verify that the calibration matches the active imaging region

Why Calibration Status Matters

Regularly reviewing calibration status helps users:

- Confirm the correct calibration is loaded before experiments
 - Detect accidental use of outdated or incorrect calibrations
 - Validate grid dimensions before running automated scans
 - Ensure ROI-dependent experiments remain spatially consistent
-

16.3 Creating Calibration Backups

Purpose of Calibration Backups

Calibration backups provide a mechanism to preserve known-good system states and protect against:

- Accidental overwriting of calibration data
- Progressive calibration drift
- Hardware or optical realignment changes
- Experimental configuration errors

Backup Creation Workflow

To create a backup:

1. The user selects the **Backup Current Calibration** option.
2. A descriptive backup name is requested (a timestamp-based default is provided).
3. The system saves a snapshot of the current calibration.

Each backup contains:

- The full galvo calibration matrix
- Grid dimensions
- Calibration type identifier
- ROI parameters (if applicable)
- Camera resolution context
- Timestamp and user-defined backup label

Storage and Organization

Backups are saved in the same calibration directory as the active calibration files. Each backup is stored independently and does not overwrite existing data.

Benefits of this approach:

- Unlimited backups
 - Clear traceability between experiments and calibrations
 - Safe rollback to previous system states
 - Long-term archival of calibration history
-

16.4 Restoring Previous Calibrations

Restore Interface

The Calibration Management window provides a **Restore Backup** function that:

- Displays all available backup names
- Avoids duplicate entries by grouping file formats
- Prevents accidental restoration of non-calibration files

Restore Priority Logic

When restoring a calibration:

1. The system first attempts to load the binary (.pkl) backup.
2. If unavailable, the JSON (.json) backup is used instead.

This ensures maximum numerical accuracy while maintaining robustness.

What Happens During Restoration

Upon successful restoration:

- The selected calibration matrix replaces the active calibration
- Grid dimensions and calibration type are updated
- The restored calibration is immediately re-saved as the active calibration
- GUI indicators and status displays are refreshed

Practical Use Cases

Restoration allows users to:

- Revert to a known-good experimental setup
- Compare performance across different calibration sessions
- Recover from accidental miscalibration
- Maintain consistency across long-term studies

16.5 Calibration File Formats (JSON, PKL)

Binary Format (.PKL)

Primary operational format

Characteristics:

- Stores NumPy arrays without conversion

- Preserves full floating-point precision
- Faster to load and save
- Preferred for automatic loading and active use

Recommended for:

- Daily operation
- High-precision experiments
- Frequent calibration switching

JSON Format (.JSON)

Human-readable and portable format

Characteristics:

- Text-based and easily inspectable
- Compatible with version control systems
- Can be opened and reviewed outside the software
- Converts arrays into standard lists

Recommended for:

- Documentation and auditing
- Debugging calibration data
- Sharing calibration metadata
- Long-term archival

Dual-Format Strategy

By saving calibration data in both formats, the system ensures:

- Maximum reliability
- Transparency for users
- Backward and forward compatibility
- Protection against file corruption

16.6 Best Practices for Calibration Maintenance

Calibration Discipline

- Recalibrate after any optical, mechanical, or alignment change
- Maintain consistent grid sizes within an experimental series
- Avoid mixing ROI and full-frame calibrations without explicit labeling

Backup Strategy

- Create backups before critical experiments
- Use descriptive, experiment-specific backup names
- Maintain at least one “baseline” or reference calibration

Verification and Validation

- Periodically verify calibration accuracy using grid scanning
- Visually inspect alignment at known reference points
- Compare restored calibrations against trusted baselines

File Management

- Do not manually edit PKL files
- Edit JSON files only if fully familiar with the data structure
- Keep calibration directories organized and, if possible, version-controlled

Hardware Awareness

Calibration is hardware-specific. Recalibrate after:

- Galvo replacement or servicing
- Optical path changes
- Camera resolution or ROI configuration changes

16.7 Summary

Calibration Management provides the infrastructure that preserves spatial accuracy across sessions, experiments, and system changes. Through automatic saving and loading, clear status reporting, robust backup and restore mechanisms, and dual-format file storage, the system ensures calibration integrity and reproducibility.

Proper calibration management is essential for reliable galvo control, accurate spatial targeting, and long-term experimental success.

End of Chapter 16

Chapter 17 – Multi-Capture Imaging System

The Multi-Capture Imaging System provides a structured framework for acquiring image data across multiple exposure settings, frame counts, and experimental conditions in a single automated run. It is designed to support quantitative imaging, signal characterization, noise analysis, and stimulation-based experiments while maintaining strict consistency in acquisition parameters. This chapter explains the rationale behind multi-capture acquisition, how exposures and frames are configured, how data is categorized, how multi-capture integrates with galvo scanning, and how progress and interruptions are handled.

17.1 Purpose of Multi-Capture

Why Multi-Capture Is Needed

Single-exposure imaging is often insufficient for advanced imaging experiments due to:

- Large dynamic range in signal intensity
- Exposure-dependent noise characteristics
- Variability in illumination or stimulation response
- The need for averaging across multiple frames

Multi-Capture addresses these limitations by allowing the system to acquire **multiple sets of images**, each with controlled exposure and frame parameters, within one coordinated workflow.

Key Objectives

The Multi-Capture Imaging System is designed to:

- Systematically explore exposure-dependent signal behavior
- Improve signal-to-noise ratio through frame averaging
- Enable fair comparison across experimental conditions
- Automate repetitive acquisition tasks
- Reduce user error by enforcing consistent parameter application

Multi-Capture is particularly valuable in experiments where repeatability, documentation, and quantitative analysis are critical.

17.2 Exposure Series Configuration

Exposure List Definition

The user defines a **series of exposure times** that the system will iterate through during acquisition. Each exposure value represents a distinct capture condition.

Exposure values:

- Are entered in microseconds or milliseconds (depending on camera configuration)
- Are applied uniformly across all frames within that exposure group
- Can be linear or logarithmically spaced

Exposure Sets

An exposure series may consist of:

- A small number of discrete exposure values for targeted testing
- A wide range of exposures to characterize system response
- Experiment-specific exposure sets optimized for signal visibility

Each exposure value defines a **capture block**, within which frames are acquired sequentially before moving to the next exposure.

17.3 Automatic Logarithmic Exposure Generation

Motivation for Logarithmic Spacing

In many imaging systems, signal intensity and noise scale non-linearly with exposure time. Logarithmic spacing allows:

- Dense sampling at low exposures where noise dominates
- Efficient coverage of high-exposure regimes
- Balanced exploration of the full dynamic range

Automatic Generation

The system supports **automatic logarithmic exposure generation**, where:

- A minimum exposure is specified
- A maximum exposure is specified
- The number of exposure steps is defined

The software computes intermediate exposures using logarithmic spacing and populates the exposure list automatically.

Advantages

- Eliminates manual calculation errors

- Ensures consistent spacing across experiments
 - Enables reproducible exposure series generation
 - Ideal for calibration, sensitivity analysis, and dynamic range studies
-

17.4 Frames per Exposure

Frame Count Configuration

For each exposure value, the user specifies the **number of frames to capture**. This defines how many images are acquired before the system moves to the next exposure.

Purpose of Multiple Frames

Capturing multiple frames per exposure enables:

- Noise reduction through averaging
- Statistical analysis (mean, variance, temporal fluctuations)
- Detection of transient artifacts
- Robust comparison across exposures

Execution Behavior

For each exposure:

1. The camera exposure is set
2. Frame acquisition begins
3. Frames are captured sequentially
4. Each frame is stored independently
5. The system proceeds to the next exposure after all frames are acquired

Frame count remains fixed across all exposures unless explicitly reconfigured.

17.5 Data Type Selection

The Multi-Capture system supports structured categorization of acquired data. Each capture run is labeled with a **data type**, ensuring clarity during analysis and documentation.

Dark Data

Purpose: Characterize sensor noise and background offset
Captured when:

- Illumination or stimulation sources are disabled
- The system is optically blocked or dark

Uses:

- Dark noise subtraction
- Offset correction
- Baseline noise characterization

Baseline Data

Purpose: Capture system response under normal, unstimulated conditions

Captured when:

- Illumination is active
- No experimental stimulation is applied

Uses:

- Reference measurements
- Normalization
- Comparison against stimulated conditions

Stimulation Data

Purpose: Capture response during active stimulation

Captured when:

- Optical, electrical, or other stimulation sources are enabled
- Galvo targeting or patterned stimulation may be active

Uses:

- Response mapping
- Differential analysis
- Functional imaging

Operational Significance

Data type labels are embedded in:

- File naming conventions
- Directory structures
- Acquisition logs

This ensures that datasets remain unambiguous and analysis-ready.

17.6 Multi-Capture with Galvo Scanning

Integration with Galvo Grid Scanning

Multi-Capture can be combined with galvo grid scanning to perform **spatially resolved, multi-exposure imaging**.

For each galvo grid point:

1. Galvos move to the target coordinate
2. Stabilization delays are applied
3. The full exposure series is executed
4. All frames for each exposure are captured
5. The system moves to the next grid point

Hierarchical Acquisition Structure

Data is organized hierarchically:

- Grid point → Exposure → Frame

This structure enables:

- Spatial response mapping
- Exposure-dependent spatial analysis
- Reproducible scanning protocols

Advantages

- Eliminates manual repositioning
- Guarantees identical acquisition parameters across space
- Enables large-scale automated experiments

17.7 Progress Monitoring and Abort

Progress Tracking

During Multi-Capture acquisition, the system provides real-time feedback including:

- Current exposure index
- Current frame number
- Total frames completed
- Overall acquisition progress

This feedback helps users:

- Estimate remaining acquisition time
- Monitor system activity
- Detect abnormal delays or issues

Abort and Safety Handling

The user may abort a Multi-Capture run at any time:

- Acquisition stops cleanly after the current frame
- No corrupted files are produced
- Partial datasets remain valid and usable

Abort functionality is essential for:

- Hardware safety
 - Experimental intervention
 - Parameter correction mid-run
-

17.8 Typical Experimental Workflows

Noise Characterization Workflow

1. Select **Dark Data**
2. Configure logarithmic exposure series
3. Capture multiple frames per exposure
4. Analyze noise scaling with exposure

Baseline vs Stimulation Comparison

1. Capture **Baseline Data** using Multi-Capture
2. Enable stimulation
3. Capture **Stimulation Data** with identical settings
4. Perform differential analysis

Spatial Response Mapping

1. Load galvo calibration
2. Enable grid scanning
3. Configure Multi-Capture exposures and frames
4. Acquire data across all grid points
5. Reconstruct spatial response maps

System Validation Workflow

1. Perform baseline Multi-Capture
 2. Restore a known calibration
 3. Repeat acquisition
 4. Compare datasets for drift or instability
-

17.9 Summary

The Multi-Capture Imaging System enables comprehensive, automated acquisition across exposure levels, frame counts, spatial positions, and experimental conditions. By combining structured exposure series, robust data typing, frame-based averaging, galvo integration, and real-time progress control, the system provides a powerful foundation for quantitative and reproducible imaging experiments.

This chapter completes the transition from calibration and scanning control to full experimental data acquisition, enabling advanced imaging workflows with minimal manual intervention.

End of Chapter 17

Chapter 18 – Data Management & File Structure

Data Management defines how all experimental outputs—including images, stacks, speckle data, calibration files, and logs—are organized, named, stored, and protected. A clear and consistent file structure is essential for traceability, reproducibility, post-processing, and long-term archival. This chapter describes how the system structures data at the session level, which file formats are used, how calibration and metadata are stored, how logs and reports are generated, and which backup strategies are recommended.

18.1 Session Directory Organization

Session-Based Data Model

All data generated by the system is organized into **session directories**. A session corresponds to a single experimental run or continuous acquisition period initiated by the user.

Each session directory is:

- Created automatically at acquisition start
- Timestamped to ensure uniqueness
- Self-contained, holding all data required to interpret the experiment

Top-Level Session Structure

A typical session directory follows this structure:

```
Session_YYYYMMDD_HHMMSS/  
├── images/  
├── stacks/  
├── speckle/  
├── calibration/  
├── logs/  
└── reports/
```

This separation ensures:

- Logical grouping of data types
- Easy navigation and filtering
- Safe copying and archival of entire experiments

Advantages of Session Isolation

- Prevents accidental mixing of datasets
- Preserves experimental context

- Enables direct session-level backup and restoration
 - Simplifies downstream analysis workflows
-

18.2 Image File Formats

Primary Image Format

Single-frame images are saved using a **lossless image format** (typically PNG or TIFF), ensuring that raw pixel values are preserved without compression artifacts.

Key characteristics:

- Full bit-depth retention
- No lossy compression
- Compatible with scientific image analysis tools

Naming Convention

Image filenames encode essential acquisition context, such as:

```
<data_type>_exp<exposure>_frame<index>_<timestamp>.png
```

When galvo scanning is enabled:

```
<data_type>_r<row>_c<col>_exp<exposure>_frame<index>.png
```

This naming scheme allows:

- Immediate identification of exposure and frame
- Direct mapping to galvo grid locations
- Chronological sorting by default

Usage Considerations

- Images are saved independently to avoid corruption
 - Partial acquisitions remain valid
 - Files can be inspected without specialized software
-

18.3 Speckle and Stack Data Formats

Stack Data (Multi-Frame Storage)

When multiple frames are acquired per exposure or per grid point, the system can generate **stack data files**.

Stack formats typically include:

- Multi-dimensional NumPy arrays
- Serialized binary formats (e.g., `.npy` or `.pkl`)

Dimensions commonly follow:

```
[frames, height, width]
```

or, in galvo-scanning contexts:

```
[grid_point, exposure, frame, height, width]
```

Speckle Data

Speckle data represents processed or derived information from raw images, such as:

- Temporal speckle contrast
- Spatial speckle statistics
- Averaged or normalized speckle maps

Speckle files:

- Are stored separately from raw images
- Preserve numerical precision
- Can be regenerated if raw data is available

Benefits of Structured Stack Storage

- Faster post-processing
- Reduced file clutter
- Direct compatibility with analysis pipelines
- Clear separation between raw and processed data

18.4 Calibration Files and Metadata

Calibration Directory

Each session contains a dedicated `calibration/` folder that stores:

- The active calibration used during acquisition
- Optional copies of backup or restored calibrations
- Metadata linking calibration to the session

Calibration Content

Stored calibration data includes:

- Galvo calibration matrices
- Grid dimensions
- Calibration method identifiers
- ROI offsets and dimensions
- Camera resolution context

Metadata Association

Calibration metadata ensures:

- Exact spatial mapping can be reconstructed
- Analysis software can interpret galvo positions correctly
- Experiments remain reproducible even months later

Calibration files are never mixed with image data, reinforcing the separation between **measurement data** and **system configuration**.

18.5 Log Files and Reports

Log Files

The system generates structured logs during acquisition and scanning, stored in the `logs/` directory.

Logs typically include:

- Acquisition start and end times
- Exposure values and frame counts
- Galvo grid traversal records
- File save confirmations
- Errors, warnings, or abort events

Logs are stored as:

- Plain text files for transparency
- Timestamped entries for traceability

Reports

In addition to logs, the system generates **human-readable reports**, often in document formats (e.g., `.docx`).

Reports may include:

- Galvo coordinate tables
- Scan summaries
- Image file references

- Calibration identifiers

Purpose of Logs and Reports

- Provide an audit trail of experiments
 - Support troubleshooting and debugging
 - Enable inclusion in experimental documentation
 - Facilitate collaboration and review
-

18.6 Recommended Backup Practices

Session-Level Backups

The preferred backup unit is the **entire session directory**. Because all data is self-contained, copying the session folder preserves the complete experiment.

Recommended practices:

- Backup immediately after acquisition
- Avoid modifying session contents post-backup
- Maintain original directory names

Redundancy Strategy

Adopt a multi-layered backup approach:

- Local backup on a secondary drive
- External storage (USB or NAS)
- Off-site or cloud archival (if permitted)

Calibration-Specific Backups

In addition to session backups:

- Maintain a separate archive of baseline calibrations
- Label calibration backups clearly
- Store calibration archives independently from experiment data

Long-Term Archival

For long-term storage:

- Compress entire session directories
- Include readme files describing experiments
- Retain logs and reports alongside data

Best Practices Summary

- Never rely on a single copy of critical data
- Treat calibration files as irreplaceable assets
- Preserve raw data even if processed data exists
- Document backup locations and dates

18.7 Summary

Data Management and File Structure provide the foundation for reliable experimentation, reproducible analysis, and long-term data integrity. By organizing all outputs into well-defined session directories, using lossless and structured file formats, preserving calibration and metadata, generating comprehensive logs and reports, and following disciplined backup practices, the system ensures that experimental data remains usable, traceable, and secure throughout its lifecycle.

This chapter completes the operational documentation by defining how experimental results are stored, protected, and prepared for analysis and archival.

End of Chapter 18

Chapter 19 – External Device Control

19.1 Relay Control System

The relay control system provides a generalized interface for switching **external hardware devices** on and off under software control. Relays are commonly used to control lasers, illumination sources, shutters, valves, or auxiliary electronics.

19.1.1 Purpose of Relay Control

Relay control allows the imaging system to:

- Enable or disable external devices programmatically
- Synchronize device states with imaging or stimulation
- Reduce manual intervention during experiments
- Improve repeatability and safety

Relays act as electrically isolated switches, protecting the control computer from high voltages or currents.

19.1.2 Relay Hardware Architecture

The relay system typically consists of:

- A relay board (USB, serial, or DAQ-controlled)
- One or more electromechanical or solid-state relays
- External power and load connections

Each relay channel corresponds to a discrete output controlled by the GUI.

19.1.3 Relay Control in the GUI

Within the GUI:

- Each relay is represented by a labeled toggle or button
- Status indicators show ON/OFF state
- Commands are issued instantly upon user interaction

Relay actions may be logged for traceability during automated experiments.

19.1.4 Safety Considerations

- Relays default to a safe OFF state at startup
 - Relay activation may be disabled if calibration or system state is invalid
 - Users should verify wiring and load ratings before operation
-

19.2 Z-Axis / Stereotaxic Stage Control

The system supports control of **Z-axis positioning devices**, including motorized focus drives and stereotaxic stages.

19.2.1 Purpose of Z-Axis Control

Z-axis control enables:

- Precise focus adjustment
- Depth positioning in biological samples
- Repeatable vertical positioning across sessions
- Integration with stereotaxic coordinates

This is especially important in in-vivo or layered sample imaging.

19.2.2 Supported Z-Axis Devices

Commonly supported devices include:

- Stepper motor-driven linear stages
- Motorized microscope focus units
- Stereotaxic manipulators with digital control

Communication is typically via serial, USB, or controller APIs.

19.2.3 Z-Axis Control Interface

The GUI provides:

- Incremental up/down controls
- Absolute position entry (if supported)
- Current position readout

- Step size configuration

Movements are executed smoothly to avoid disturbing the sample.

19.2.4 Coordinate Systems and Reference

Z-axis position may be reported in:

- Motor steps
- Micrometers or millimeters
- Relative offsets from a user-defined zero

Users should establish a consistent reference position at the start of experiments.

19.2.5 Safety and Limits

- Software-defined travel limits prevent overrun
 - Emergency stop functionality halts motion immediately
 - Motion is disabled during unsafe system states
-

19.3 NI-DAQ Stimulation Controller

The NI-DAQ stimulation controller provides **precise timing and signal generation** for experimental stimulation.

19.3.1 Role of NI-DAQ in the System

The NI-DAQ module is used to:

- Generate digital or analog stimulation signals
- Trigger external stimulators
- Synchronize stimulation with imaging frames
- Record timing-critical events

It acts as the central timing backbone for stimulation experiments.

19.3.2 Supported Output Types

Typical outputs include:

- Digital TTL pulses
- Analog voltage waveforms
- Trigger signals for external devices

Each output channel can be configured independently.

19.3.3 DAQ Initialization and Status

Before use:

- The DAQ device must be detected and initialized
- Channel availability is verified
- Error states are reported in the GUI

DAQ status indicators help prevent silent failures.

19.4 Stimulation Parameter Configuration

Stimulation parameters define **how, when, and for how long** external stimulation is applied.

19.4.1 Core Stimulation Parameters

Typical configurable parameters include:

- Pulse amplitude
- Pulse width
- Pulse frequency
- Number of pulses
- Inter-pulse interval
- Total stimulation duration

These parameters determine the physiological or experimental effect.

19.4.2 GUI-Based Configuration

The stimulation control panel allows:

- Numeric entry of parameters
- Validation of acceptable ranges
- Preview of timing sequences
- Saving and reloading stimulation presets

Preset management ensures consistency across experiments.

19.4.3 Single vs Repetitive Stimulation

The system supports:

- Single-shot stimulation
- Repetitive or burst-based stimulation
- Trial-based stimulation sequences

Users should clearly document the chosen mode.

19.4.4 Safety and Validation

- Parameter limits prevent unsafe outputs
 - Invalid configurations are blocked
 - Dry-run testing without output is recommended before live experiments
-

19.5 Synchronization with Imaging

Synchronization ensures that **imaging data and external events share a common time reference**.

19.5.1 Importance of Synchronization

Precise synchronization allows:

- Correlation of physiological changes with stimulation
- Frame-accurate event marking
- Reliable temporal analysis

Without synchronization, interpretation of results becomes ambiguous.

19.5.2 Triggering Modes

Common synchronization modes include:

- Imaging-triggered stimulation
- Stimulation-triggered imaging

- Shared clock or trigger source

The selected mode depends on experimental requirements.

19.5.3 Synchronization Workflow

A typical synchronized experiment involves:

1. Configuring stimulation parameters
2. Enabling synchronization mode
3. Starting live or speckle imaging
4. Initiating stimulation via DAQ
5. Capturing synchronized imaging data

Timestamps and markers are stored with the imaging data.

19.5.4 Handling Latency and Jitter

The system minimizes timing errors by:

- Using hardware-based triggers
- Avoiding software polling where possible
- Logging precise trigger times

Users should account for any residual latency in analysis.

19.6 Summary

This chapter has provided a **complete overview of external device control**, including relay systems, Z-axis and stereotaxic control, NI-DAQ-based stimulation, parameter configuration, and synchronization with imaging. Proper integration and disciplined operation of these components enable **complex, reproducible, and temporally precise experiments** within the imaging platform.

End of Chapter 19

Chapter 20 – Performance Optimization, Automated and Batch Experiment Workflows

20.1 Overview and Objectives

As experiments grow in complexity and duration, manual operation becomes inefficient and error-prone. This chapter addresses how to:

- Optimize system performance for sustained operation
- Minimize latency and dropped frames
- Automate repetitive imaging and stimulation tasks
- Execute batch experiments with consistent parameters
- Ensure reproducibility across long experimental sessions

The tools and strategies described here are essential for **high-throughput imaging**, **longitudinal studies**, and **multi-condition experiments**.

20.2 Performance Optimization Fundamentals

Performance optimization ensures that the system operates at its intended speed and accuracy without instability.

20.2.1 Camera and Acquisition Optimization

Key considerations include:

- **Resolution and ROI**
Use the smallest ROI that fully contains the region of interest. Smaller ROIs reduce data throughput and increase achievable frame rates.
 - **Exposure Time**
Avoid unnecessarily long exposure times. Exposure should be optimized for signal quality without saturating pixels.
 - **Preview vs Acquisition Mode**
Disable live preview when running long automated acquisitions to free processing resources.
-

20.2.2 Speckle Processing Performance

Speckle computation is computationally intensive. Performance can be optimized by:

- Choosing an appropriate frames-per-speckle value
- Avoiding excessively large ROIs
- Reducing speckle video refresh rate when real-time display is not required
- Disabling visualization overlays during batch runs

Balancing temporal resolution and computational load is critical.

20.2.3 Disk I/O and Data Storage

High-speed imaging generates large volumes of data.

Best practices include:

- Saving data to SSDs rather than HDDs
 - Using dedicated storage drives for experiments
 - Avoiding network drives during acquisition
 - Monitoring available disk space before starting batch runs
-

20.2.4 System Resource Management

To maintain stable performance:

- Close unnecessary background applications
 - Avoid CPU- or GPU-intensive tasks during experiments
 - Ensure adequate system cooling
 - Monitor memory usage during long sessions
-

20.3 Principles of Automation in Imaging Experiments

Automation enables precise, repeatable control over complex experimental sequences.

20.3.1 Why Automate Experiments

Automation allows:

- Consistent execution of repeated protocols
- Reduced user intervention and fatigue
- Improved timing accuracy
- Scalable multi-condition testing

Automated workflows are particularly valuable for comparative and longitudinal studies.

20.3.2 Building Blocks of Automation

Automated experiments are composed of discrete actions, such as:

- Camera initialization
- Exposure and ROI configuration
- Galvo positioning
- External device triggering
- Image acquisition and saving
- Data labeling and logging

These actions are executed in a predefined order.

20.4 Automated Imaging Workflows

20.4.1 Single-Condition Automated Acquisition

A basic automated workflow may include:

1. Initialize camera and verify calibration
2. Set exposure and ROI
3. Start image or speckle acquisition
4. Save data with timestamped filenames
5. Stop acquisition and release resources

This workflow ensures consistency across repeated trials.

20.4.2 Multi-Exposure Automation

For experiments requiring multiple exposure settings:

- A predefined list of exposure times is iterated
- At each exposure:
 - The camera is reconfigured
 - Stabilization delay is applied
 - Acquisition is performed
 - Data is saved with exposure-specific identifiers

This approach is useful for dynamic range studies and optimization experiments.

20.4.3 ROI and Galvo-Based Automation

Automated workflows may also include:

- Sequential galvo targeting of predefined coordinates
- ROI repositioning or resizing
- Spatial scanning protocols

Each spatial condition is treated as a separate acquisition block.

20.5 Batch Experiment Design

Batch experiments execute **multiple automated workflows sequentially** without user intervention.

20.5.1 Defining a Batch Experiment

A batch experiment consists of:

- A list of experimental conditions
- Parameter sets for each condition
- Execution order and repetition count

Conditions may vary by exposure, ROI, galvo position, stimulation parameters, or time.

20.5.2 Batch Configuration Interface

The GUI allows users to:

- Define parameter matrices
- Enable or disable specific conditions
- Set iteration counts
- Specify delays between runs

Clear labeling of batch parameters is essential for later analysis.

20.5.3 Execution and Monitoring

During batch execution:

- Progress indicators show current condition and iteration
- Logs record timestamps and system states
- Errors trigger safe termination or user alerts

Users should monitor early stages of batch runs to confirm correct behavior.

20.6 Automated Stimulation and Imaging Integration

Automation is particularly powerful when combined with external device control.

20.6.1 Coordinated Stimulation Protocols

Automated workflows may include:

- Pre-imaging baseline acquisition
- Timed stimulation via NI-DAQ
- Post-stimulation imaging
- Repeated stimulation-imaging cycles

Timing parameters are defined explicitly to ensure reproducibility.

20.6.2 Synchronization in Automated Runs

Synchronization is maintained by:

- Hardware triggers
- Shared clocks
- Timestamped event markers

Automation ensures that synchronization settings are applied consistently across all trials.

20.7 Error Handling and Fail-Safe Behavior

Long automated runs must anticipate and handle errors gracefully.

20.7.1 Common Failure Scenarios

- Camera disconnection

- Disk space exhaustion
 - Invalid calibration state
 - External device communication failure
-

20.7.2 Automated Responses

The system may:

- Pause execution
- Safely stop acquisition
- Disable galvo or stimulation outputs
- Save partial data and logs

Clear error reporting helps users diagnose issues quickly.

20.8 Data Organization and Metadata Management

Automated and batch workflows generate large datasets that must be organized carefully.

20.8.1 File Naming Conventions

Recommended practices:

- Include experiment ID, condition, and iteration number
 - Encode exposure, ROI, or stimulation parameters
 - Use consistent, machine-readable formats
-

20.8.2 Metadata Logging

Each automated run should log:

- All imaging parameters
- Calibration references
- Timing and synchronization details
- System status and errors

Metadata is essential for downstream analysis and publication.

20.9 Summary

This chapter has detailed how to **optimize system performance**, **design automated imaging workflows**, and **execute large-scale batch experiments** reliably. By combining careful performance tuning with structured automation, users can achieve high-throughput, reproducible imaging with minimal manual intervention, even in complex, multi-parameter experimental designs.

End of Chapter 20

Chapter 21 – Troubleshooting, Diagnostics, and Safety Guidelines

21.1 System Diagnostics and Health Checks

21.1.1 Startup Diagnostics

Perform these checks when launching the system:

- COM Port Detection:**
[PORT DETECTION] Available COM ports: ['COM3', 'COM4', 'COM5']
[PORT DETECTION] RELAY_GALVO found on COM4
[PORT DETECTION] XYZ_STAGE not found by description
[PORT DETECTION] LASER not found by description
 - Expected:** At least COM4 detected for relay/galvo
 - Action if Missing:** Check USB connections, device power
- Camera Initialization:**
DEBUG: Camera process started
DEBUG: Initializing camera
DEBUG: Camera initialized successfully
 - Expected:** "Camera initialized successfully"
 - Error Indicators:** "Initialization failed" or timeout
- Calibration Load Status:**
[Calibration] Loaded 3x3 regression calibration
 - Expected:** Previous calibration loaded if available
 - Warning:** "No saved calibration found" (normal on first use)

21.1.2 Real-Time Status Monitoring

Monitor these indicators during operation:

<i>Status Location</i>	<i>Normal Indicator</i>	<i>Warning Indicator</i>	<i>Critical Indicator</i>
Header Laser	"--- mA" (stable)	Fluctuating values	"--- mA" (no reading)

<i>Status Location</i>	<i>Normal Indicator</i>	<i>Warning Indicator</i>	<i>Critical Indicator</i>
Current			
Bottom Status Bar	"Ready"	Orange text warnings	Red text errors
Galvo Motor Status	"Motor X: 0.00°"	"Motor X: --"	"ERROR" in text
ROI Status	"ROI: ACTIVE"	"ROI: INVALID"	"ROI: ERROR"
Capture Progress	Smooth progression	Paused at value	Reset to 0 unexpectedly

21.2 Troubleshooting by Symptom

21.2.1 Camera and Imaging Issues

Problem: "No live frame available"

Possible Causes:

1. Camera not initialized
2. Preview not started
3. Camera disconnected during operation
4. Driver conflict

Diagnostic Steps:

1. Check camera process status in console:
 - DEBUG: Camera process started ✓
 - DEBUG: Camera initialized successfully ✓
 - DEBUG: Displaying live frame ✓
2. Verify physical connections
3. Restart camera:
 - Stop Preview → Initialize Camera → Start Preview

Problem: Flickering or unstable live feed

Causes:

1. Exposure time too low
2. USB bandwidth issues
3. Lighting fluctuations
4. Software conflicts

Solutions:

1. Increase exposure time (5000+ μ s for testing)
2. Use single USB 3.0 port exclusively
3. Disable other camera software
4. Check for fluorescent light interference (use 50Hz/60Hz avoidance)

Problem: Speckle contrast not updating

Causes:

1. Speckle video not started
2. FPS set too high for processing
3. ROI changed during operation
4. Frame buffer mismatch

Solutions:

1. Check speckle video is running (button shows "Stop Speckle Video")
2. Reduce FPS to 5-10
3. Restart speckle video after ROI changes
4. Clear frame buffer: Stop → Wait 2s → Start

21.2.2 Galvo Control Issues

Problem: Galvo not responding to commands

Diagnostic Checklist:

1. **Galvo Enabled?** (Button shows "Disable Galvo" in green)
2. **COM Connection?** Check console for Modbus errors
3. **Power Supply?** Verify galvo controller power LED
4. **Calibration Loaded?** Status shows matrix available

Console Debug Output:

[Galvo X] ☒ Sent angle 10.50 on attempt 1 # SUCCESS
[Galvo X] Attempt 1: Error sending angle 10.50 # FAILURE
[Galvo X] ☒ Failed to move after retries. # TIMEOUT

Problem: Calibration points misaligned

Causes:

1. ROI changed after calibration
2. Manual angle entry errors
3. Galvo mechanical drift
4. Incorrect point clicking order

Recalibration Procedure:

1. **Reset:** Clear current calibration in Management window
2. **Verify ROI:** Disable or keep consistent
3. **Manual Check:** Use single point verification:

Send Angle to Motor 1 (X): 0.0°
Send Angle to Motor 2 (Y): 0.0°
Verify beam position matches expected center
4. **Recalibrate:** Use hybrid method with manual verification

Problem: Coordinate scan skipping points

Causes:

1. Matrix index mismatch
2. Communication delays
3. Frame capture failures

Debug Mode:

1. Enable verbose logging in coordinate selector:

[Galvo Scan] Starting coordinate scan...
[Galvo Scan] Moving to point 1/9 — Row:1 Col:1 — X:10.50 Y:-5.20
[Galvo Scan] Captured and saved frame at (row 0, col 0)
2. Check file saving in target directory
3. Verify galvo movement delays (2s between axes)

21.2.3 Data Acquisition Problems

Problem: Multi-capture stopping unexpectedly

Error Investigation:

1. Check console for error messages
2. Review partial saved files

3. Verify disk space availability
4. Check stimulation conflicts if enabled

Recovery Options:

1. **Resume:** If paused, use Resume Capture
2. **Continue from last exposure:** Note last saved file timestamp
3. **Restart with different parameters:** Reduce frame count or exposures

Problem: Saved files corrupted or incomplete

File Validation:

1. **MAT Files:** Load in MATLAB/Python to verify structure
2. **Images:** Check file sizes (16-bit PNG > 1MB typically)
3. **Metadata:** Review DOCX logs for completion entries

Prevention:

1. Ensure stable power supply during capture
2. Use SSD storage instead of network drives
3. Close other applications during long captures
4. Implement save verification in scripts

21.2.4 Communication Errors

Problem: "Modbus Error" or "No response"

Serial Communication Debug:

1. **Port Availability:**

```
# Test in Python console
import serial.tools.list_ports
ports = list(serial.tools.list_ports.comports())
print([p.device for p in ports])
```

2. **Baud Rate Verification:** Ensure 9600 matches device
3. **Cable Testing:** Try different USB-serial cable
4. **Device Reset:** Power cycle the galvo/relay controller

Problem: Laser current monitor showing "--- mA"

Diagnosis:

1. Check serial connection to laser controller
2. Verify baud rate (usually 9600 or 115200)
3. Test with serial terminal software (Putty, RealTerm)

4. Check laser power supply and safety interlocks

21.3 Diagnostic Tools and Utilities

21.3.1 Built-in Diagnostic Functions

Calibration Status Report

Access via **Calibration Management** → **Refresh Status**

✓ CALIBRATION LOADED

Grid: 3×3

Type: regression

Shape: (3, 3, 2)

File Status:

- JSON file: ✓ Found
- Binary file: ✓ Found
- Backup files: 3 found

ROI Settings (if applicable):

- Active: True
 - Region: (100,100) to (1820,1340)
-

System Health Check

Create custom diagnostic script:

```
def system_diagnostic():
    checks = {
        'Camera Process': check_camera_process(),
        'Modbus Connection': test_modbus_com4(),
        'Calibration Loaded': check_calibration_file(),
        'Disk Space': check_disk_space('C:', 5), # 5GB minimum
        'Python Dependencies': verify_dependencies(),
        'USB Bandwidth': estimate_usb_throughput()
    }
```

return checks

21.3.2 Log File Analysis

System generates several log sources:

1. **Console Log:** Real-time Python print statements
2. **Error Log:** `traceback.format_exc()` outputs
3. **Operation Log:** `coordinate_log.docx`
4. **Performance Log:** Timing arrays in memory

Typical Error Patterns:

Communication Timeout

"Timeout waiting for response" - Increase delays in galvo control

Memory Error

"MemoryError: Unable to allocate array" - Reduce frame buffer size

File Access Conflict

"PermissionError: [Errno 13]" - Close other programs accessing files

21.3.3 Performance Monitoring

Timing Benchmarks

Monitor during operation:

In multi-capture function

```
timing_data = {  
    'galvo_move': galvo_move_times,    # Should be ~2.0s per axis  
    'exposure_capture': exposure_capture_times, # Depends on exposure  
    'image_save': image_save_times,    # Should be <1.0s per image  
    'total_cycle': capture_times      # Sum of above  
}
```

Acceptable Ranges:

- Galvo Movement: 1.5-2.5s per axis
 - Image Save: <2.0s for MAT file
 - Speckle Calculation: <0.1s per frame at 10 FPS
-

21.4 Safety Guidelines

21.4.1 Laser Safety

Class 3B/4 Laser Protocols

1. **Eye Protection:**
 - Wear appropriate OD-rated goggles at all times
 - Match wavelength (650nm default) to goggle specification
 - Ensure goggles cover entire beam path area
2. **Beam Enclosure:**
 - Fully enclose beam path when possible
 - Use beam stops and shutters
 - Install "Laser On" indicator lights
3. **Interlock Systems:**
 - Door interlocks on experimental enclosure
 - Emergency stop buttons accessible
 - Key-controlled laser activation

Laser Current Monitoring Safety

- **Warning Threshold:** >1000 mA requires immediate shutdown
- **Stability Requirement:** Fluctuations <5% during operation
- **Emergency Procedure:** If monitoring fails, assume laser is ON

21.4.2 Electrical Safety

Modbus/Device Connections

1. **Grounding Verification:**
 - All equipment on common ground
 - Verify isolation between power and signal grounds
 - Use opto-isolators for long cable runs
2. **Current Limits:**
 - Galvo motors: Check rated current (typically 2A max)
 - Relay loads: Do not exceed contact ratings (usually 5A)
 - Power supplies: 20% overhead on rated capacity
3. **Cable Management:**
 - Secure cables to prevent tripping hazards
 - Label all power and signal cables
 - Use strain relief on connectors

21.4.3 Stimulation Safety

NIDAQ Stimulation Protocols

Pre-Stimulation Checklist:

- ☐ 1. Stimulus parameters verified by second person
- ☐ 2. Electrode impedance checked ($<50\text{k}\Omega$)
- ☐ 3. Animal physiological monitoring active
- ☐ 4. Emergency stop accessible
- ☐ 5. Stimulation isolation verified

Parameter Safety Limits:

- **Pulse Duration:** Maximum 1.0ms (current: 0.3ms)
- **Current/Voltage:** Below tissue damage thresholds
- **Frequency:** Below seizure induction limits for model

Multi-Capture with Stimulation

Safety Interlocks:

1. Stimulation only when explicitly enabled
2. Visual confirmation required before start
3. Abort capture immediately stops stimulation
4. Post-stimulation monitoring period enforced

21.4.4 Biological Specimen Safety

During Imaging Sessions

1. **Anesthesia Monitoring:**
 - Continuous physiological parameter tracking
 - Temperature maintenance system
 - Respiratory rate monitoring
 2. **Positioning and Stability:**
 - Secure stereotaxic mounting
 - Pressure point protection
 - Regular position checks during long scans
 3. **Recovery Protocols:**
 - Post-experimental observation period
 - Analgesia administration as needed
 - Veterinary consultation for adverse events
-

21.5 Emergency Procedures

21.5.1 Immediate Shutdown Scenarios

Situation 1: Laser runaway (increasing current)

1. HIT EMERGENCY STOP (physical button)
2. Disconnect laser power supply
3. Close laser shutter if available
4. Notify all personnel in area
5. DO NOT look at beam path

Situation 2: Electrical smoke or smell

1. Cut main power to experiment
2. Use appropriate fire extinguisher (Class C for electrical)
3. Evacuate if smoke persists
4. Contact facilities maintenance

Situation 3: Software freeze during stimulation

1. Hardware abort: Disconnect stimulator power
2. Software kill: Task Manager → End Python process
3. Animal care: Immediate physiological assessment
4. System reboot required before reuse

21.5.2 Controlled Shutdown Procedure

Normal Operation Termination:

1. Stop all acquisition: [Stop Capture] [Stop Preview]
2. Return galvo to home position: [Return to Start]
3. Disable devices: [Disable Galvo] → Turn gray
4. Save current state: Manual backup of calibration
5. Close GUI: [Exit GUI] button (not window close)
6. Power down hardware in reverse startup order

21.5.3 Data Recovery After Crash

If system crashes during acquisition:

1. **Check last saved files:** Timestamp indicates progress
 2. **Recover partial MAT files:** May contain partial data
 3. **Review error logs:** %TEMP% folder for Python tracebacks
 4. **Calibration backup:** Auto-saved after each calibration
-

21.6 Preventive Maintenance

21.6.1 Daily Checks

1. **Visual Inspection:**
 - Laser beam alignment
 - Cable integrity
 - Cooling system operation
 - Enclosure interlocks
2. **Software Verification:**
 - Calibration loading test
 - Single exposure capture
 - Galvo movement test (small angles)
3. **Data Integrity:**
 - Verify save directory writable
 - Check disk space (>10GB free)
 - Test file saving speed

21.6.2 Weekly Maintenance

1. **Hardware Connections:**
 - Tighten all connectors
 - Clean optical surfaces
 - Check for mechanical wear
2. **Software Updates:**
 - Backup calibration files
 - Review error logs
 - Update dependencies if stable
3. **Performance Validation:**
 - Full calibration workflow
 - Multi-capture dry run (no saving)
 - Stimulation system test (no subject)

21.6.3 Quarterly Calibration

1. **Optical System:**
 - Laser power calibration
 - Camera flat-field correction
 - Spatial calibration verification
 2. **Mechanical Systems:**
 - Galvo angle calibration against reference
 - Stage position accuracy check
 - Bearing and movement smoothness
 3. **Electrical Systems:**
 - Ground continuity verification
 - Current measurement calibration
 - Stimulation output calibration
-

21.7 Regulatory Compliance

21.7.1 Required Documentation

Maintain these records:

1. **Laser Safety:** Training records, inspection logs
2. **Animal Protocols:** IACUC approvals, procedure logs
3. **Electrical Safety:** Inspection certificates
4. **Software Validation:** Version control, change logs

21.7.2 Warning Labels and Signage

Required in experiment area:

1. **Laser Warning Sign:** Class, wavelength, maximum power
2. **Electrical Hazard Signs:** High voltage locations
3. **Emergency Procedures:** Posted at each station
4. **Contact Information:** PI and safety officer contacts

21.7.3 Training Requirements

Mandatory for all users:

1. Laser safety training (annual)
2. Animal handling certification (if applicable)
3. Electrical safety awareness
4. Software operation proficiency test
5. Emergency procedure drills (biannual)

21.8 Quick Reference: Error Codes and Solutions

<i>Error Message</i>	<i>Likely Cause</i>	<i>Immediate Action</i>	<i>Long-term Solution</i>
"Camera initialization failed"	Driver conflict, busy	Restart computer	Reinstall scTDC drivers
"No response from galvo"	COM port disconnect	Check USB cable	Implement heartbeat monitoring
"ROI coordinates invalid"	$X2 \leq X1$ or $Y2 \leq Y1$	Reset ROI to defaults	Add input validation
"MemoryError"	Too many frames in buffer	Reduce FPS/frame count	Implement streaming to disk
"Calibration matrix mismatch"	ROI changed after cal	Recalibrate with current ROI	Store ROI with calibration
"Stimulation timeout"	NIDAQ communication fail	Check NIDAQ connection	Add hardware watchdog
"File save permission denied"	File in use by other app	Close other programs	Implement file locking

Remember: When in doubt, **STOP** the procedure, **ASSESS** the situation, and **SEEK HELP** if safety is uncertain. Never bypass safety interlocks or override safety warnings. Regular maintenance and adherence to protocols prevent most issues before they occur.

Safety First, Data Quality Second, Speed Third.

End of Chapter 21

Chapter 22 – Quick References and Appendices

22.1 Typical End-to-End Workflow

22.1.1 Full Experimental Session (2-3 hours)

Phase 1: PREPARATION (30 minutes)

- |— 1.1 System Power-Up
 - | |— Turn on laser power supply
 - | |— Power galvo controller
 - | |— Start camera cooling (if applicable)
 - | |— Power computer and displays
- |— 1.2 Software Launch
 - | |— Run new_GUI_sc_55.py
 - | |— Verify COM port detection
 - | |— Confirm calibration loaded
- |— 1.3 Hardware Check
 - | |— Laser beam alignment
 - | |— Galvo range test ($\pm 10^\circ$)
 - | |— Camera focus adjustment
 - | |— Stimulation electrode placement
- |— 1.4 Subject Preparation
 - | |— Anesthesia induction
 - | |— Stereotaxic positioning
 - | |— Physiological monitoring start

Phase 2: CALIBRATION (20 minutes)

- |— 2.1 Scale Calibration
 - | |— Click "Calibrate Scale"
 - | |— Draw line between known marks
 - | |— Enter 1.0mm (or known distance)
- |— 2.2 ROI Setup
 - | |— Camera Controls → ROI Settings
 - | |— Set region of interest
 - | |— Enable ROI
- |— 2.3 Galvo Calibration
 - | |— Click "Multi-Capture & Galvo"
 - | |— Select 3×3 grid

- | | — Click "Auto Calibrate"
- | | — Complete 9-point calibration
- | — 2.4 Verification
 - | — Test home position
 - | — Scan single point
 - | — Save calibration backup

Phase 3: BASELINE ACQUISITION (30 minutes)

- | — 3.1 Multi-Capture Setup
 - | | — Set exposures: 500,1000,1500,2000,2500 μ s
 - | | — Frames: 100
 - | | — Check "Collect Baseline Images"
 - | | — Select save directory
- | — 3.2 Start Acquisition
 - | | — Click "Start Capture"
 - | | — Monitor progress
 - | | — Verify file saving
- | — 3.3 Quality Check
 - | — Review first speckle image
 - | — Check intensity profiles
 - | — Verify galvo positions

Phase 4: STIMULATION EXPERIMENT (60+ minutes)

- | — 4.1 Stimulation Setup
 - | | — Open NiDAQ Controller
 - | | — Set parameters: 3 sets, 90 pulses
 - | | — Enable stimulation in Multi-Capture
 - | | — Safety verification
- | — 4.2 Timed Acquisition
 - | | — Pre-stimulation: 2 minutes baseline
 - | | — During stimulation: Start capture → Start stimulation
 - | | — Post-stimulation: 5 minutes recovery
 - | | — Repeat for multiple conditions
- | — 4.3 Real-Time Monitoring
 - | — Watch laser current
 - | — Monitor speckle contrast changes
 - | — Check subject physiology

Phase 5: ANALYSIS & SHUTDOWN (30 minutes)

- | — 5.1 Immediate Analysis
 - | | — Save speckle images
 - | | — Export intensity profiles

- | └─ Generate quick-look plots
- | └─ 5.2 Data Organization
- | └─ Rename files with experiment ID
- | └─ Create README with parameters
- | └─ Backup to secure storage
- | └─ 5.3 System Shutdown
- └─ Return galvo to home
- └─ Stop all acquisition
- └─ Close GUI properly
- └─ Power down hardware in reverse order

22.1.2 Quick Measurement (15 minutes)

1. Launch GUI (already calibrated)
2. Start Preview → Check image
3. Enable ROI if needed
4. Start Speckle Video (10 FPS)
5. Adjust C-axis slider for contrast
6. Draw intensity profile line
7. Save speckle image
8. Export MAT file for analysis

22.1.3 Calibration-Only Session

1. System check (no subject)
2. Scale calibration with calibration slide
3. Galvo calibration (hybrid method):
 - 3×3 grid
 - Manual angles for corners
 - Dial-based for center
 - Click all 9 points
4. Test scan pattern
5. Save calibration with descriptive name
6. Create backup

22.2 Default Settings Summary

22.2.1 Camera and Imaging Defaults

<i>Parameter</i>	<i>Default Value</i>	<i>Range</i>	<i>Unit</i>
Exposure Time	1500	100-10000	μs
Frames for Speckle Video	10	2-100	frames
Speckle Video FPS	10	1-30	Hz
Live Contrast	1.000	0.000-1.000	factor
Speckle C-axis	1.000	0.000-1.000	1/k ²
Camera Resolution	1920×1440	Fixed	pixels

22.2.2 ROI Defaults

<i>Parameter</i>	<i>Default Value</i>	<i>Max Value</i>
X1 (left)	0	1919
Y1 (top)	0	1439
X2 (right)	1920	1920
Y2 (bottom)	1440	1440
Default ROI State	Disabled	-

22.2.3 Galvo Defaults

<i>Parameter</i>	<i>Default Value</i>	<i>Range</i>	<i>Notes</i>
Motor X Range	-40.0° to +40.0°	±40.0°	Dial limits
Motor Y Range	-40.0° to +40.0°	±40.0°	Dial limits

<i>Parameter</i>	<i>Default Value</i>	<i>Range</i>	<i>Notes</i>
Scroll Steps	0.1°	Fixed	Per dial click
Default Angle	0.0°	-	Home position
Move Delay	2.0s	Fixed	Between axes
Grid Size	3×3	2×2 to 3×3	Calibration

22.2.4 Multi-Capture Defaults

<i>Parameter</i>	<i>Default Value</i>	<i>Example</i>
Exposures	"500,1000,1500,2000,2500,3000"	Comma-separated
Frames per Exposure	100	Integer
Auto-Exposure Min	500 μs	Lowest value
Auto-Exposure Max	5000 μs	Highest value
Auto-Exposure Count	5	Number of points
Data Types	None selected	Checkboxes

22.2.5 Stimulation Defaults (NiDAQ)

<i>Parameter</i>	<i>Default Value</i>	<i>Notes</i>
Number of Sets	1	Stimulation bouts
Pulses per Set	90	Per stimulation
Pulse Duration	0.3 ms	300 μs
Inter-Pulse Interval	166.4 ms	~6 Hz

<i>Parameter</i>	<i>Default Value</i>	<i>Notes</i>
Inter-Set Interval	30 s	Recovery period

22.2.6 File System Defaults

<i>Setting</i>	<i>Default Value</i>	<i>Can be Changed?</i>
Save Directory	Current working directory	Yes (Browse)
Calibration File	galvo_calibration.pkl	Auto-managed
Backup Location	Script directory	Fixed
Image Format	.png (16-bit) + .mat	Fixed
Log Format	.docx	Fixed

22.3 Mouse and Interaction Summary

22.3.1 Primary Canvas Interactions

Live Image Canvas

<i>Action</i>	<i>Effect</i>	<i>Visual Feedback</i>
Hover	Shows coordinates	Status bar + coordinates display
Click (Calibration)	Records point	None (programmatic)
Click+Drag (Intensity)	Draws profile line	Red line appears

<i>Action</i>	<i>Effect</i>	<i>Visual Feedback</i>
Click+Drag (Calibration Tool)	Draws scale line	Green line appears

Speckle Image Canvas

<i>Action</i>	<i>Effect</i>	<i>Notes</i>
Hover	No action	Coordinates not mapped
Click+Drag (Intensity)	Draws profile line	Red line appears
Right-click	No action	Reserved for future

22.3.2 Slider Controls

Live Contrast Slider

- **Location:** Left of live image (vertical)
- **Range:** 0.000 (min contrast) to 1.000 (max)
- **Steps:** 0.001 increments
- **Effect:** Adjusts display contrast (not saved)
- **Real-time:** Updates immediately

C-Axis Slider (Speckle)

- **Location:** Left of speckle image (vertical)
- **Label:** "C-Axis ($1/k^2$)"
- **Range:** 0.000 to 1.000
- **Steps:** 0.001 increments
- **Effect:** Scales speckle contrast: $\text{display} = (1/k^2) \times \text{factor}$
- **Updates:** Real-time speckle visualization

22.3.3 Button Types and States

Toggle Buttons

[Enable Galvo] → Gray (disabled)

[Disable Galvo] → Green (enabled)

[Start Preview] → Normal (available)

[Start Preview] → Disabled (running)

[Stop Preview] → Normal (when previewing)

☐ Enable ROI → Unchecked

☒ Enable ROI → Checked

State-Indicating Buttons

<i>Button Text</i>	<i>State</i>	<i>Color</i>
Relay 1 ON	Active	Green
Relay 1 OFF	Inactive	Gray
Start Capture	Ready	Default
Pause Capture	Active	Yellow (conceptual)
Stimulation: ENABLED	Ready	Green text
Stimulation: DISABLED	Off	Red text

22.3.4 Dialog Interactions

Confirmation Dialogs

Appear for:

1. Galvo movement confirmation
2. Stimulation start
3. Dark image collection warning
4. Calibration overwrite

Response Guidelines:

- **OK/Yes:** Proceed with caution
- **Cancel/No:** Abort, check settings
- **Always read** the warning message completely

Data Entry Dialogs

Types:

1. **Simple string:** Calibration backup name
2. **Numeric entry:** Real distance in mm
3. **Grid position:** "1,2" for row,col
4. **Angle entry:** "10.5,-5.2" for X,Y

Format Requirements:

- Angles: X.XX,Y.YY (comma-separated, no spaces)
 - Grid: row,col (1-indexed)
 - Distance: numeric value only
-

22.4 Common User Mistakes

22.4.1 Calibration Errors

Mistake 1: Calibrating with wrong ROI

Symptom: Points don't align when ROI enabled/disabled

Solution:

1. Decide ROI state before calibration
2. Recalibrate if ROI needs change
3. Store ROI settings with calibration

Mistake 2: Clicking wrong image location

Symptom: Poor regression fit, inconsistent mapping

Prevention:

- Zoom mentally: Identify exact beam center
- Use consistent clicking technique
- Verify each point before proceeding

Mistake 3: Mixing angle conventions

Symptom: Mirror or rotated mapping

Remember:

- Image coordinates: (0,0) = top-left
- Galvo angles: Positive/negative as labeled
- Grid order: Row-major (row, then column)

22.4.2 Acquisition Mistakes

Mistake 4: Starting capture without checking settings

Checklist before "Start Capture":

- ☐ 1. Exposure times verified
- ☐ 2. Frame count appropriate
- ☐ 3. Save directory exists and writable
- ☐ 4. Data type(s) selected
- ☐ 5. Galvo enabled/disabled as needed
- ☐ 6. Stimulation parameters verified (if used)

Mistake 5: Changing settings during acquisition

Golden Rule: Never change ROI, exposure, or galvo during capture

Exception: Pause → Change → Resume (if supported)

Mistake 6: Overwriting files

Symptom: Lost previous data, duplicate names

Prevention:

- Timestamp automatically added to filenames
- Use descriptive directory names
- Manual backup before new experiment

22.4.3 System Operation Mistakes

Mistake 7: Wrong shutdown order

Correct Order:

1. Software stop (all acquisitions)
2. Galvo home position
3. GUI close via "Exit GUI" button
4. Hardware power down (reverse of startup)

Incorrect: Closing window with X, killing Python process

Mistake 8: Ignoring warning messages

Common warnings to NEVER ignore:

1. "Laser current unstable"
2. "ROI coordinates invalid"
3. "Calibration not loaded"
4. "Disk space low"

Mistake 9: Using wrong calibration for experiment

Verification Protocol:

1. Check calibration timestamp
2. Test single point mapping
3. Compare to physical beam position
4. Recalibrate if >1mm discrepancy

22.4.4 Data Management Mistakes

Mistake 10: Poor file organization

Bad Practice: All files in one folder

Good Practice:

2024-06-15_Experiment1/

```
|— calibration/
|   |— galvo_calibration.pkl
|   |— scale_factor_15.2_ppmm.txt
|— raw_data/
|   |— dark_data/
|   |— base_data/
|   |— stim_data/
|— processed/
|   |— speckle_images/
|   |— intensity_profiles/
|— metadata/
|   |— experiment_notes.docx
|   |— parameter_log.json
```

Mistake 11: Not documenting parameters

Minimum Documentation:

1. Laser power/wavelength
2. Camera exposure settings
3. Galvo calibration grid
4. Stimulation parameters
5. Subject/condition information
6. Any deviations from protocol

22.5 Appendices

Appendix A: Configuration and Settings Files

A.1 Primary Configuration Files

1. tdc_gpx3.ini (**Camera Configuration**)

```
[Camera]
Model=GPX3
Resolution=1920x1440
BitDepth=12
FPS_Max=30
```

```
[Acquisition]
DefaultExposure=1500
DefaultGain=1.0
TriggerMode=Internal
```

```
[Save]
DefaultFormat=PNG16
Compression=None
```

Location: Same directory as script

2. galvo_calibration.json (**Human-readable**)

```
json
{
  "galvo_matrix": [[[10.5, -5.2], [10.5, 0.0], ...]],
  "grid_rows": 3,
  "grid_cols": 3,
  "matrix_type": "regression",
  "roi_offset_x": 0,
  "roi_offset_y": 0,
  "roi_width": 1920,
  "roi_height": 1440,
  "timestamp": "2024-06-15T14:30:22",
  "camera_resolution": [1920, 1440]
}
```

3. galvo_calibration.pkl (**Binary backup**)

- Same data as JSON but with numpy arrays preserved
- Used by system for quick loading

A.2 Auto-generated Files

During operation:

1. coordinate_log.docx: Galvo scan positions
2. *_stats.docx: Intensity profile statistics
3. calibration_backup_*.pkl: Timestamped backups

A.3 User-customizable Settings

Not in config files (GUI-only):

- Default save directory
- Recent exposure lists
- Manual angle entries
- ROI coordinates (unless saved with calibration)

Appendix B: Mathematical Definitions

B.1 Speckle Contrast Calculation

Raw speckle contrast (per pixel over time):

Let $I(t)$ = intensity at time t for N frames

Mean: $\mu = (1/N) * \sum I(t)$

Standard deviation: $\sigma = \sqrt{(1/N) * \sum (I(t) - \mu)^2}$

Raw contrast: $K = \sigma / \mu$

Display speckle contrast (in GUI):

With C-axis factor f (0.000 to 1.000):

Display value = $1 / K^2 * f$

Where K^2 is the speckle contrast squared

Why $1/K^2$?: Common in LSCI literature for blood flow relation

B.2 Calibration Regression

For N calibration points:

Image coordinates: (x_i, y_i) for $i = 1..N$

Galvo angles: $(\theta_{x_i}, \theta_{y_i})$

Linear regression models:

$$\theta_x = a_0 + a_1 * x + a_2 * y \text{ (fit to all } \theta_{x_i})$$

$$\theta_y = b_0 + b_1 * x + b_2 * y \text{ (fit to all } \theta_{y_i})$$

Grid generation from regression:

For grid of size R×C:

$$x_grid = \text{linspace}(\min(x), \max(x), C)$$

$$y_grid = \text{linspace}(\min(y), \max(y), R)$$

For each (y, x) in grid:

$$\theta_{x_pred} = a_0 + a_1 * x + a_2 * y$$

$$\theta_{y_pred} = b_0 + b_1 * x + b_2 * y$$

Result: R×C×2 matrix

B.3 ROI Coordinate Transformations

Without ROI:

Display coordinates → Image coordinates:

$$\text{img_x} = (\text{disp_x} - \text{pad_left}) / \text{scale}$$

$$\text{img_y} = (\text{disp_y} - \text{pad_top}) / \text{scale}$$

With ROI active:

ROI coordinates:

$$\text{roi_x} = \text{img_x}$$

$$\text{roi_y} = \text{img_y}$$

Original image coordinates:

$$\text{orig_x} = \text{roi_x} + \text{roi_x1}$$

$$\text{orig_y} = \text{roi_y} + \text{roi_y1}$$

B.4 Temporal Sampling

Speckle video buffer:

FPS = frames per second

Buffer_size = FPS frames

Update_rate = 1/FPS seconds

Contrast calculation uses all frames in buffer

Multi-capture timing:

Total_time = exposures × frames × (exposure + overhead)

Where overhead ≈ 0.1s per frame (saving, transfer)

Appendix C: File Naming Conventions

C.1 Auto-generated Names

Pattern: {type}_{description}_{timestamp}_{suffix}.{ext}

Examples:

live_image_15062024_143022.png
speckle_image_15062024_143022.mat
raw_stack_15062024_143022.mat
preview_speckle_15062024_143022.png
live_intensity_profile_15062024_143022.mat
dark_data_15062024_143022.mat

C.2 Timestamp Format

DDMMYYYY_HHMMSS

Examples:

15062024_143022 = June 15, 2024, 14:30:22

01012024_090015 = Jan 1, 2024, 09:00:15

C.3 User-specified Names

Calibration backups:

User input: "pre_stim_calibration"

Becomes: pre_stim_calibration.pkl

pre_stim_calibration.json

Experiment directories:

Recommended: YYYY-MM-DD_ExperimentName_SubjectID

Example: 2024-06-15_WhiskerStim_Mouse12

C.4 File Type Summary

<i>Extension</i>	<i>Content</i>	<i>Size (approx)</i>	<i>Opens With</i>
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<i>Extension</i>	<i>Content</i>	<i>Size (approx)</i>	<i>Opens With</i>
.mat	MATLAB data	10-100 MB	MATLAB, Python (scipy.io)
.pkl	Python pickle	1-10 KB	Python (pickle.load)
.json	JSON text	5-20 KB	Any text editor
.png	16-bit image	1-5 MB	ImageJ, Photoshop, Python
.docx	Word document	10-100 KB	Microsoft Word, LibreOffice
.txt	Log/notes	1-10 KB	Any text editor

Appendix D: Glossary of Terms

D.1 Hardware Terms

<i>Term</i>	<i>Definition</i>
Galvo	Galvanometer, a motor that rotates mirrors for beam steering
LSCI	Laser Speckle Contrast Imaging, technique for blood flow visualization
Modbus	Serial communication protocol for industrial devices
NIDAQ	National Instruments Data Acquisition hardware
ROI	Region of Interest, subset of image for processing
scTDC	Specific camera model/library used in system
Stereotaxic	Precise 3D positioning system for biological specimens

D.2 Software Terms

<i>Term</i>	<i>Definition</i>
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<i>Term</i>	<i>Definition</i>
C-axis	Contrast adjustment axis ($1/k^2$ scaling)
Hybrid Calibration	Combination of manual and dial-based angle specification
MAT file	MATLAB data file format storing arrays and metadata
Multi-capture	Automated acquisition at multiple exposure times
Pickle (.pkl)	Python object serialization format
Regression Matrix	3D array mapping image coordinates to galvo angles
Speckle Contrast	Statistical measure of laser speckle pattern variation

D.3 Mathematical Terms

<i>Term</i>	<i>Definition</i>	<i>Formula/Notes</i>
Speckle Contrast (K)	Normalized standard deviation of intensity	$K = \sigma/\mu$
Temporal Contrast	Contrast computed over time at single pixel	
Spatial Contrast	Contrast computed over space at single time	Less common in this system
px/mm	Pixels per millimeter, calibration factor	Measured via scale calibration
$1/k^2$	Inverse of contrast squared	Proportional to correlation time

D.4 Process Terms

<i>Term</i>	<i>Definition</i>
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<i>Term</i>	<i>Definition</i>
Baseline Acquisition	Imaging without stimulation, reference condition
Dark Image	Image with no light, for noise characterization
Grid Scan	Systematic movement through calibration points
Home Position	Default/starting galvo position (usually 0,0)
Intensity Profile	Line plot of pixel values along drawn line
Stimulation Trial	Combined imaging and stimulation period

D.5 User Interface Terms

<i>Term</i>	<i>Definition</i>	<i>Location/Appearance</i>
Live Feed	Real-time camera image	Left display panel
Speckle Feed	Processed contrast image	Right display panel
Status Bar	Bottom text display	"Ready" or status messages
Coord Display	Coordinate readout	Camera Controls window
Dial/Meter	Circular angle selector	Galvo control section
Slider	Vertical contrast control	Beside image displays

D.6 Error/Status Terms

<i>Term</i>	<i>Meaning</i>	<i>Action Required</i>
"No response"	Modbus communication failed	Check connections, restart
"Matrix not loaded"	Calibration missing	Run calibration first
"ROI invalid"	Coordinates incorrect	X2>X1, Y2>Y1, within bounds

<i>Term</i>	<i>Meaning</i>	<i>Action Required</i>
"Previewing"	Camera actively acquiring	Normal operation
"Initialized"	Camera ready for commands	Can start preview
"Galvo enabled"	Galvo control active	Can send movement commands

Quick Index to Common Tasks

Finding Functions:

- **Camera controls:** "Open Camera Controls" button
- **Galvo calibration:** "Multi-Capture & Galvo" → Calibration section
- **Scale calibration:** "Calibrate Scale" in header
- **Stimulation control:** "NiDAQ Controller" in header
- **File management:** Save directory browser in Multi-Capture
- **Intensity tools:** Left panel under "Live/Speckle Intensity Profile"

Emergency Stops:

1. **Software freeze:** Task Manager → End Python
2. **Laser emergency:** Physical emergency stop button
3. **Stimulation emergency:** Disconnect stimulator power
4. **Galvo runaway:** Disable Galvo button

Daily Startup Checklist:

1. Power hardware in order
2. Launch GUI
3. Check calibration loaded
4. Test camera preview
5. Verify galvo movement

6. Confirm save location
7. Ready for experiment

Weekly Maintenance:

1. Back-up calibration files
2. Clean optical surfaces
3. Test full workflow
4. Review error logs
5. Update experiment log

Final Note: This system represents a sophisticated integration of optical, mechanical, and software components. Success depends on understanding both the individual components and their interactions. Regular practice with the system, careful documentation, and adherence to safety protocols will ensure reliable operation and high-quality scientific data.

End of Chapter 22

End of User Manual